



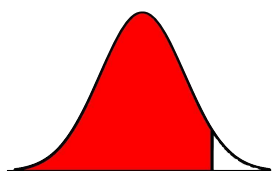
Integrating Data-Level Balancing and Enhanced RFM-Based Feature engineering for Credit Card Fraud Detection

Student Name: Andy Chen, Alvin Lin, Chrissy Wang

Advisor Name: Hui Wen Teng

Outline

1. Introduction
2. Data
3. Methodology
4. Results
5. Conclusion



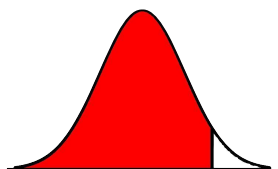
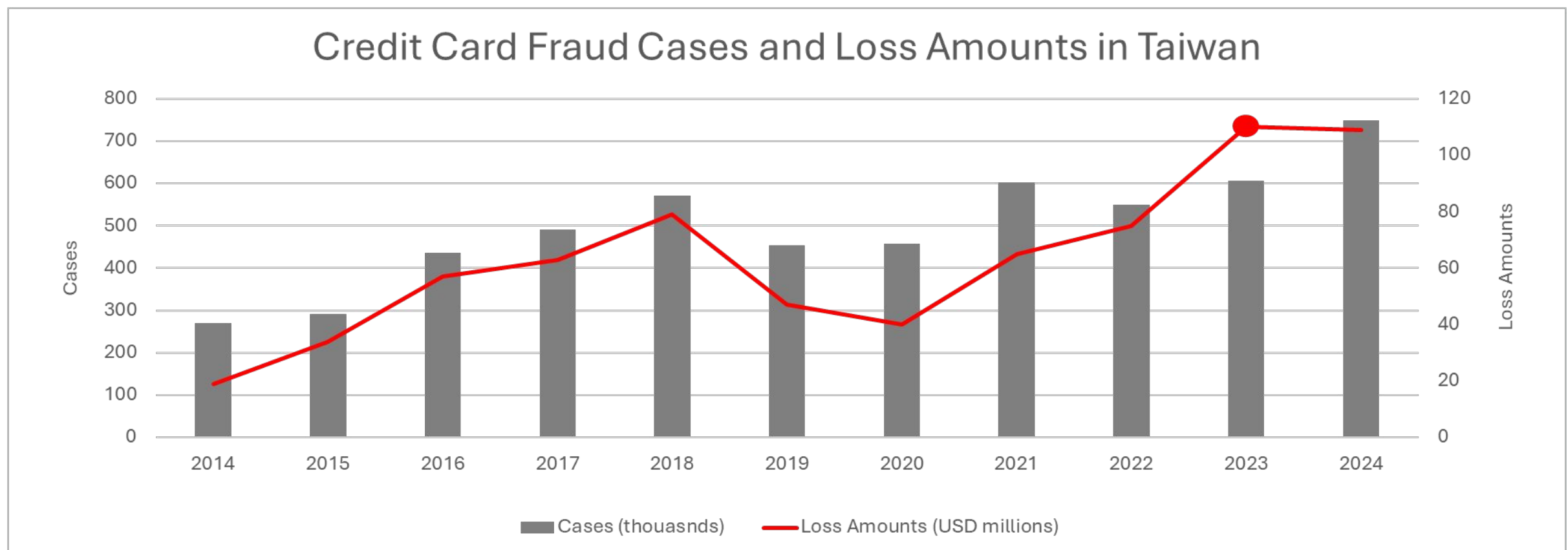
Importance of Credit Card Fraud Detection

World: Fraud losses reached US\$ 33.83 billion in 2023 and US\$33.45 billion in 2022.

United States: Losing over US\$10 billion to fraud in 2023 with a 14% increase from 2022.

Europe: Fraud totaled US\$4.9 billion in 2022.

Taiwan: Fraud losses hit US\$110 million in 2023, the highest in a decade.



Characteristics of Fraud Detection

(Bart Baesens et al., 2021)

Concealed

Carefully planned and hard to capture by fixed rules.

Feature Engineering

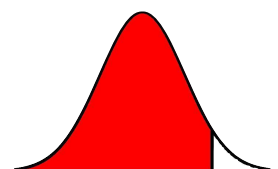
Rare

Low occurrence causes class imbalance and weak learning.

Imbalanced Learning Techniques

Evolving

Continuously changing, requiring model adaptability.

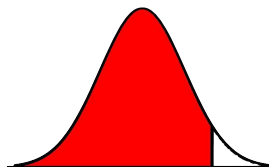


Feature engineering frameworks in fraud detection

RFM framework

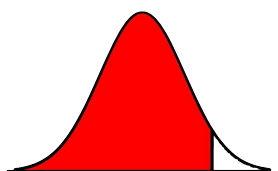
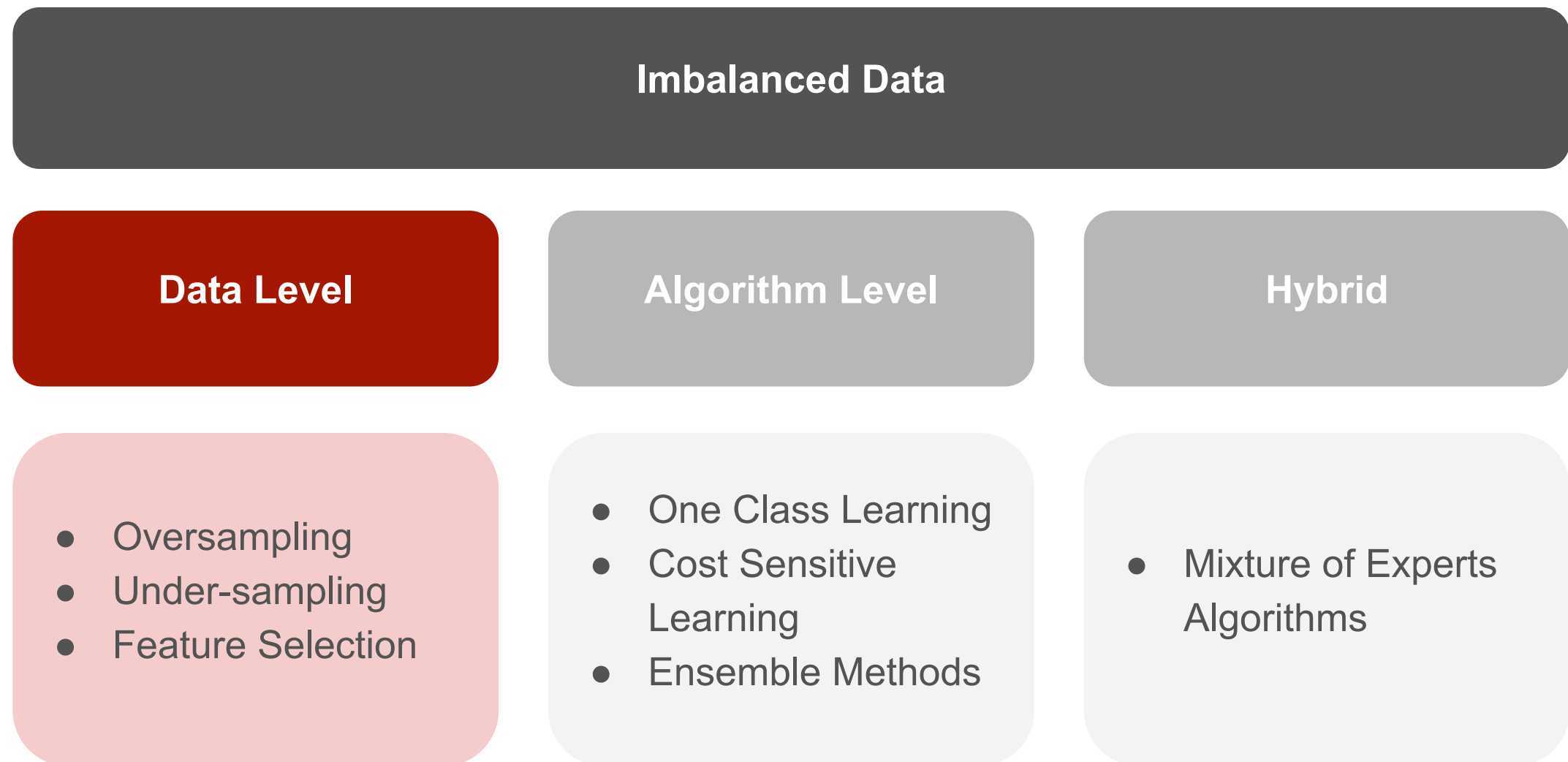
Component	What It Measures	Fraud Detection Insight
Recency	How recently a customer performed a specific type of transaction.	Long gaps in expected behavior may indicate suspicious activity.
Frequency	How often a customer makes certain types of transactions within a time window (e.g., 90/180 days).	Sudden surge or drop to zero may signal abnormal or fraudulent activity.
Monetary	The total or average amount of money spent in transactions over a given period.	Unusually high amounts can be a strong fraud indicator.

(Bart Baesens et al., 2021) (Xinwei Zhang et al., 2019)



Imbalanced Learning Techniques

(Spelman et al., 2018)



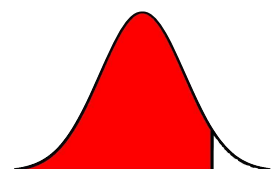
Financial Transactions Dataset: Analytics

Dataset Overview

- Source: Kaggle
- Duration: 2010/01/01~2019/10/31
- Size: 13,305,915 Observations (Only 8,914,963 have labels)

Composition

- Source datasets (5):
 - Transaction
 - User
 - Credit Card
 - Train_Fraud_Labels
 - MCC_Codes
- Columns: 37 (expanded to 51 after one-hot encoding and missing value imputation)
- Target labels: fraud_label
- Overall Fraud Rate: 0.100% (0.149% among labeled samples)



Source columns

Red: join Key | **Green: missing_flag** | **Blue: one hot encoding**

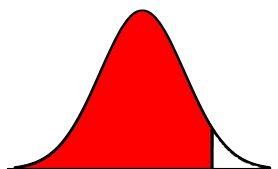
users_df	
#	Columns
0	client_id
1	current_age
2	retirement_age
3	birth_year
4	birth_month
5	gender
6	address
7	latitude
8	longitude
9	per_capita_income
10	yearly_income
11	total_debt
12	credit_score
13	num_credit_cards

transactions_df	
#	Columns
0	transaction_id
1	date
2	client_id
3	card_id
4	amount
5	use_chip
6	merchant_id
7	merchant_city
8	merchant_state
9	zip
10	mcc_code
11	errors
12	merchant_state_missing_flag
13	zip_missing_flag
14	errors_missing_flag

transactions_df	
#	Columns
0	card_id
1	client_id
2	card_brand
3	card_type
4	card_number
5	expires
6	cvv
7	has_chip
8	num_cards_issued
9	credit_limit
10	acct_open_date
11	year_pin_last_changed
12	card_on_dark_web
13	card_type_Credit
14	card_type_Debit
16	card_brand_Amex
17	card_brand_Discover
18	card_brand_Mastercard
19	card_brand_Visa

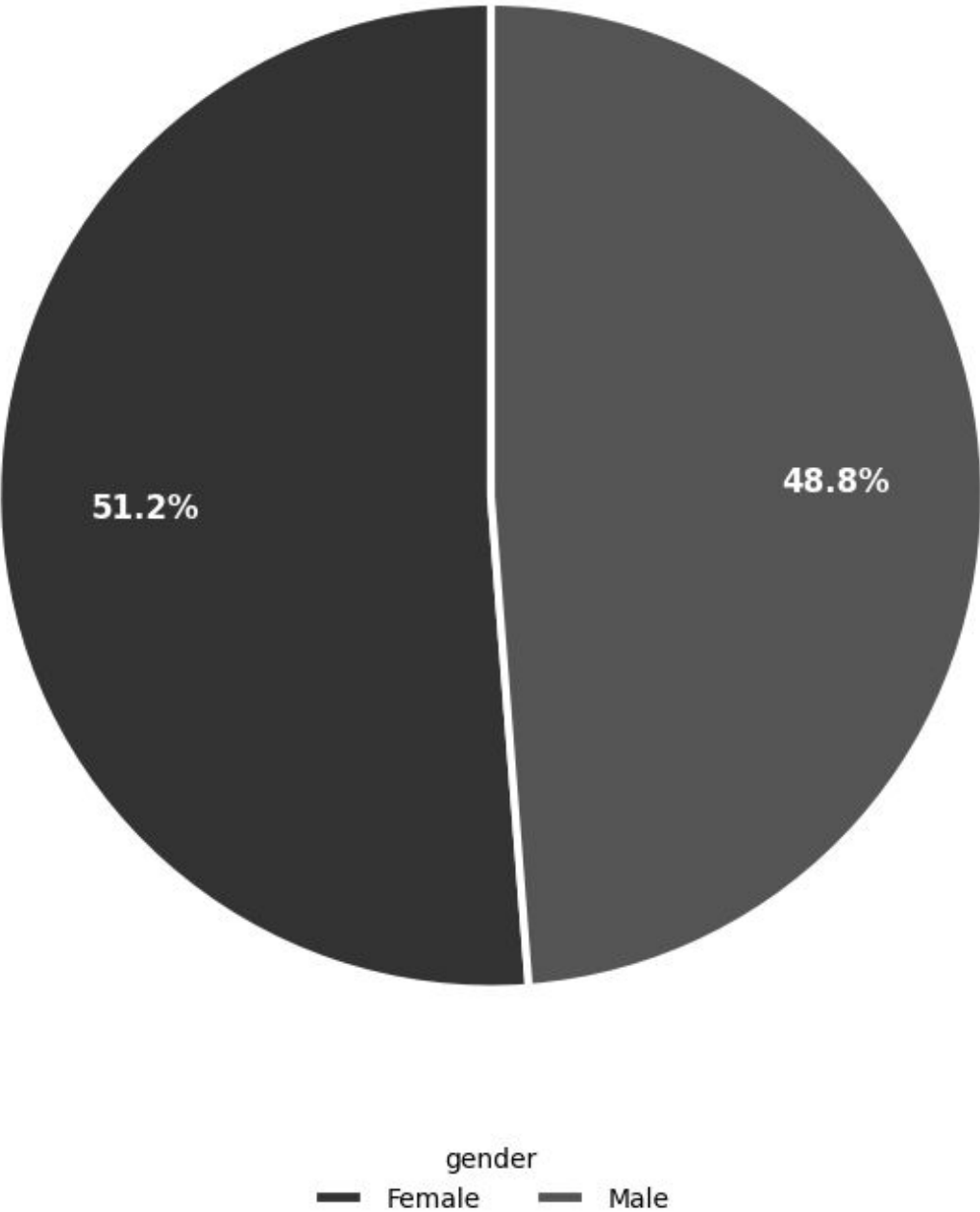
train_fraud_labels_df	
#	Columns
0	transaction_id
1	fraud_label

mcc_codes_df	
#	Columns
0	mcc_code
1	category_name

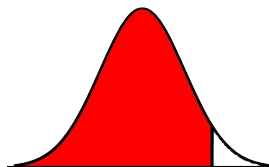


Categorical Variables—Consumers’ Gender

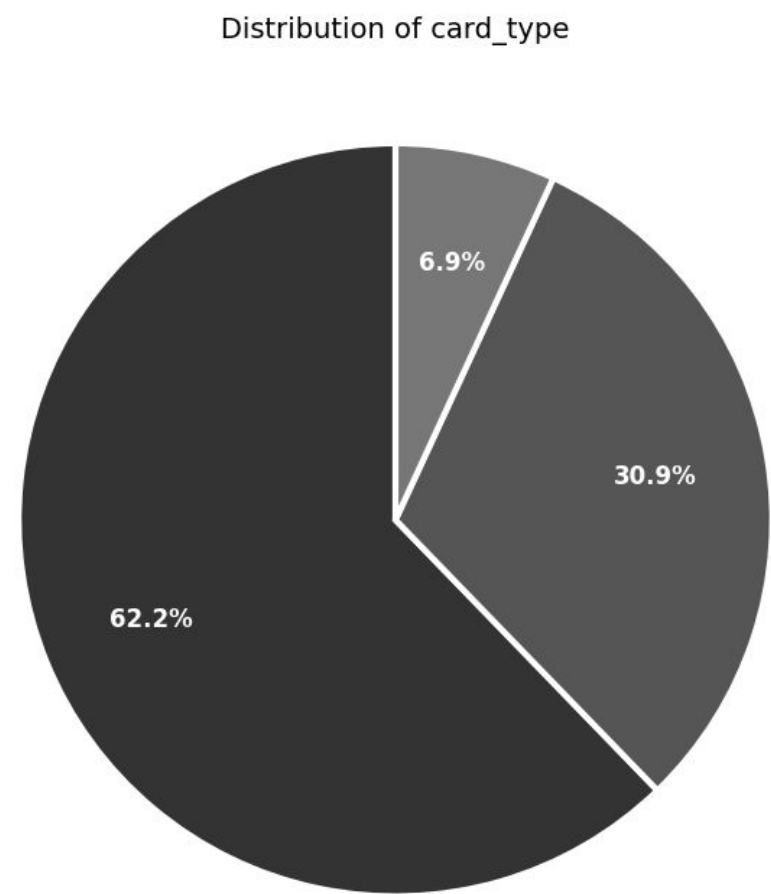
Distribution of gender



Gender	Frequency
Female	6,815,916 (51.22%)
Male	6,489,999 (48.78%)

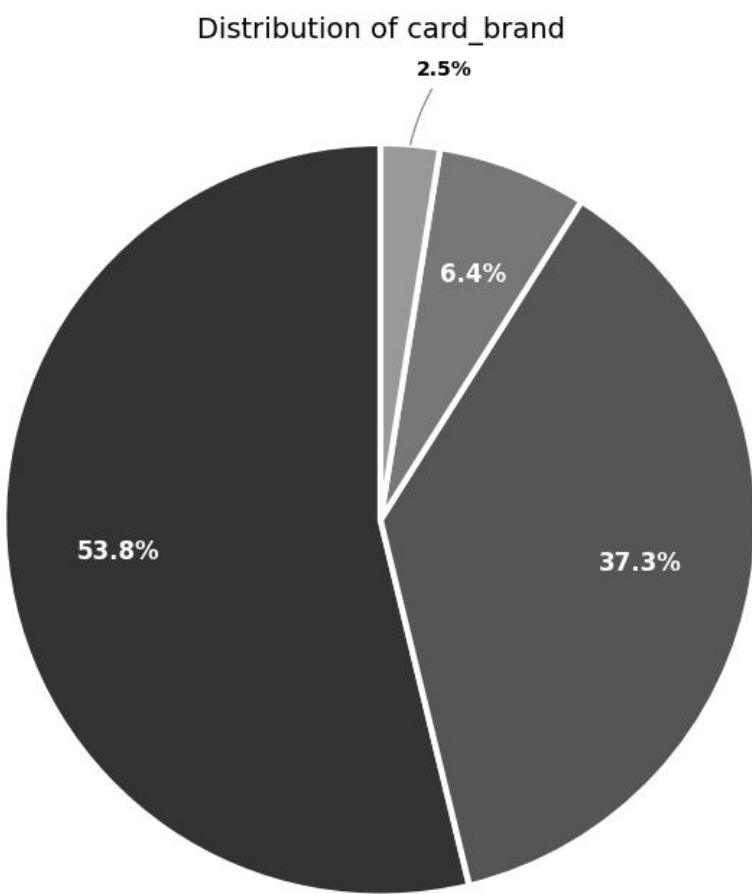


Categorical Variables—Card Information



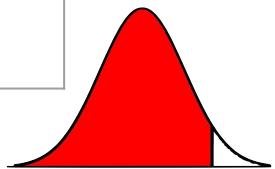
card_type
Debit Credit Debit (Prepaid)

Card Type	Frequency
Debit	8,280,996 (62.24%)
Credit	4,109,189 (30.88%)
Debit (Prepaid)	915,730 (6.88%)



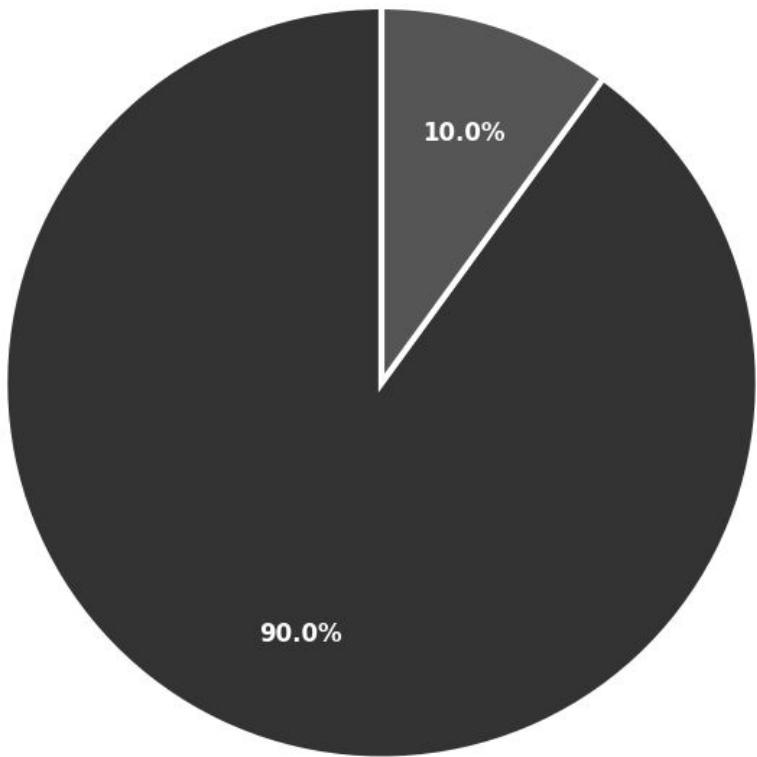
card_brand
Mastercard Visa Amex Discover

Card Brand	Frequency
Mastercard	7,157,399 (53.79%)
Visa	4,957,563 (37.26%)
Amex	854,490 (6.42%)
Discover	336,463 (2.53%)



Categorical Variables—Chip

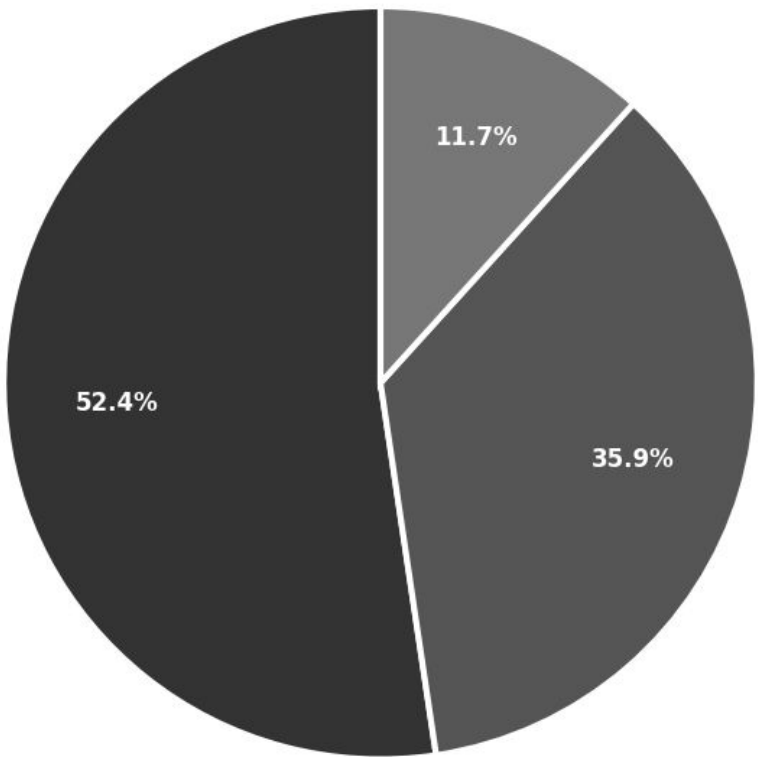
Distribution of has_chip



has_chip
1 0

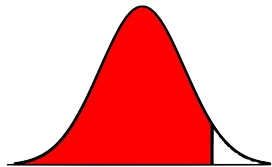
Has Chip	Frequency
Yes (1)	11,974,795 (90.00%)
No (0)	1,331,120 (10.00%)

Distribution of use_chip



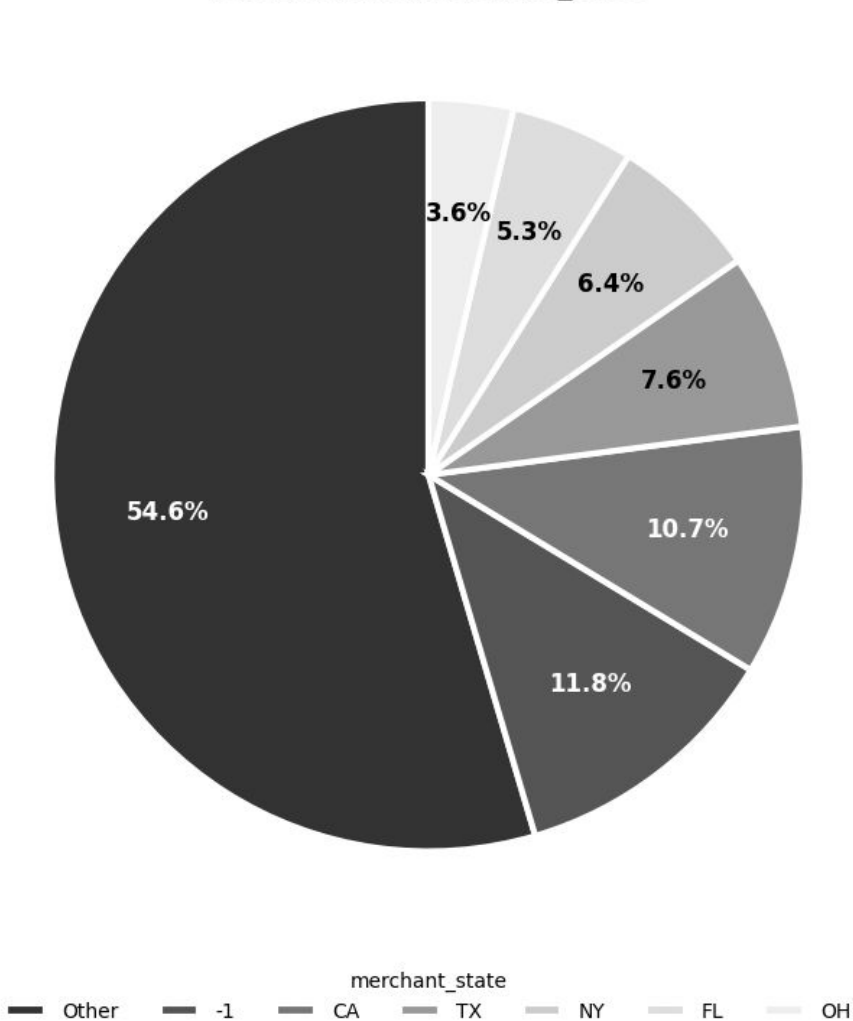
use_chip
Swipe Transaction Chip Transaction Online Transaction

Use Chip	Frequency
Swipe Transaction	6,967,185 (52.36%)
Chip Transaction	4,780,818 (35.93%)
Online Transaction	1,557,912 (11.71%)

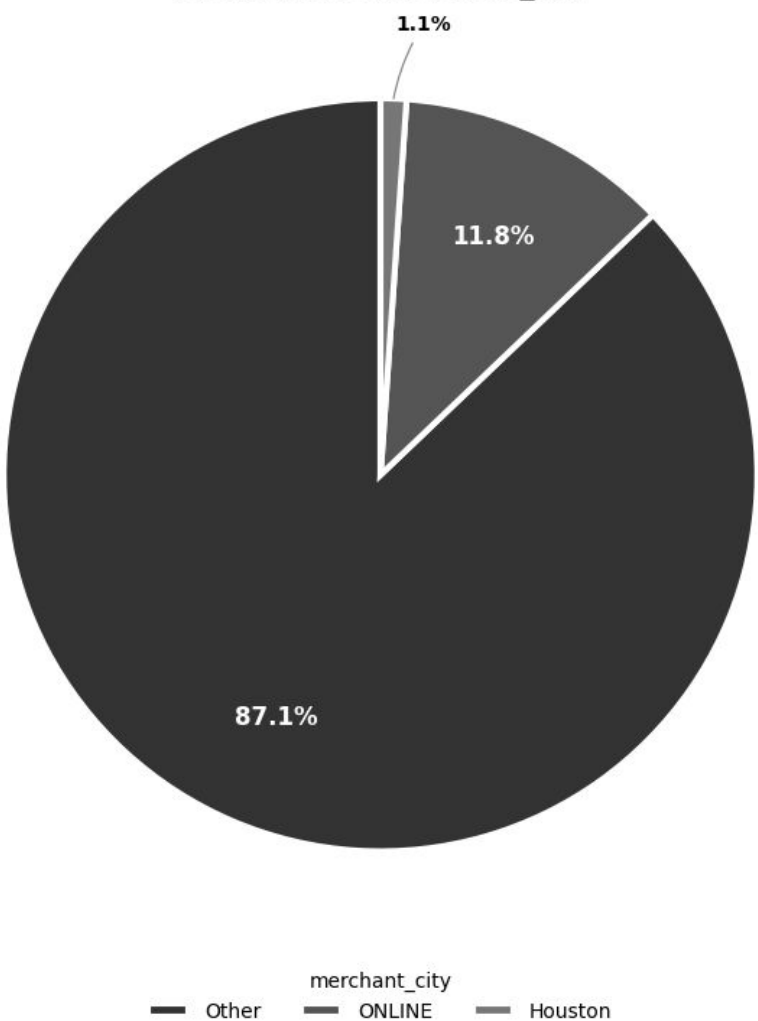


Categorical Variables—Merchant Location/Online

Distribution of merchant_state

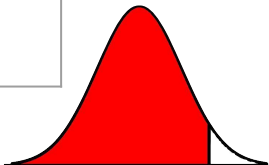


Distribution of merchant_city

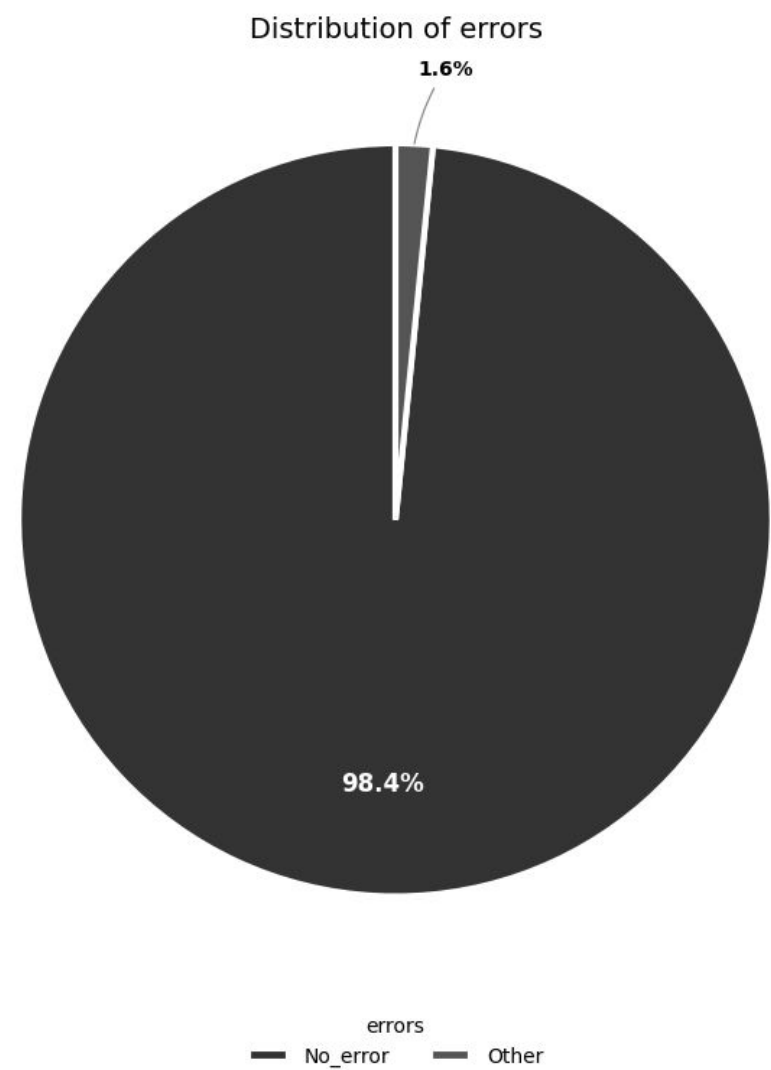


Merchant State	Frequency
Online	1,563,700 (11.75%)
CA	1,427,087 (10.73%)
TX	1,010,207 (7.59%)
NY	857,510 (6.44%)

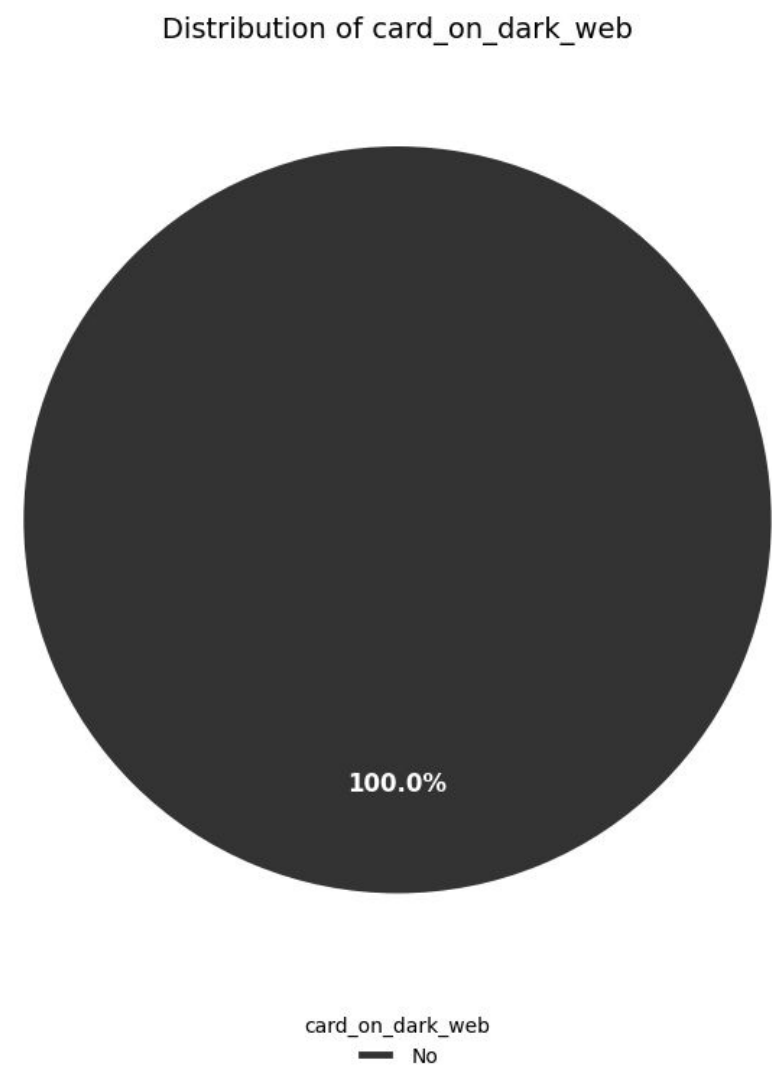
Merchant City	Frequency
Online	1,563,700 (11.75%)
Houston	146,917 (1.10%)
Miami	87,388 (0.66%)
Brooklyn	84,020 (0.63%)



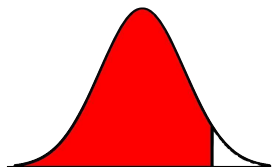
Categorical Variables—Abnormal Transaction



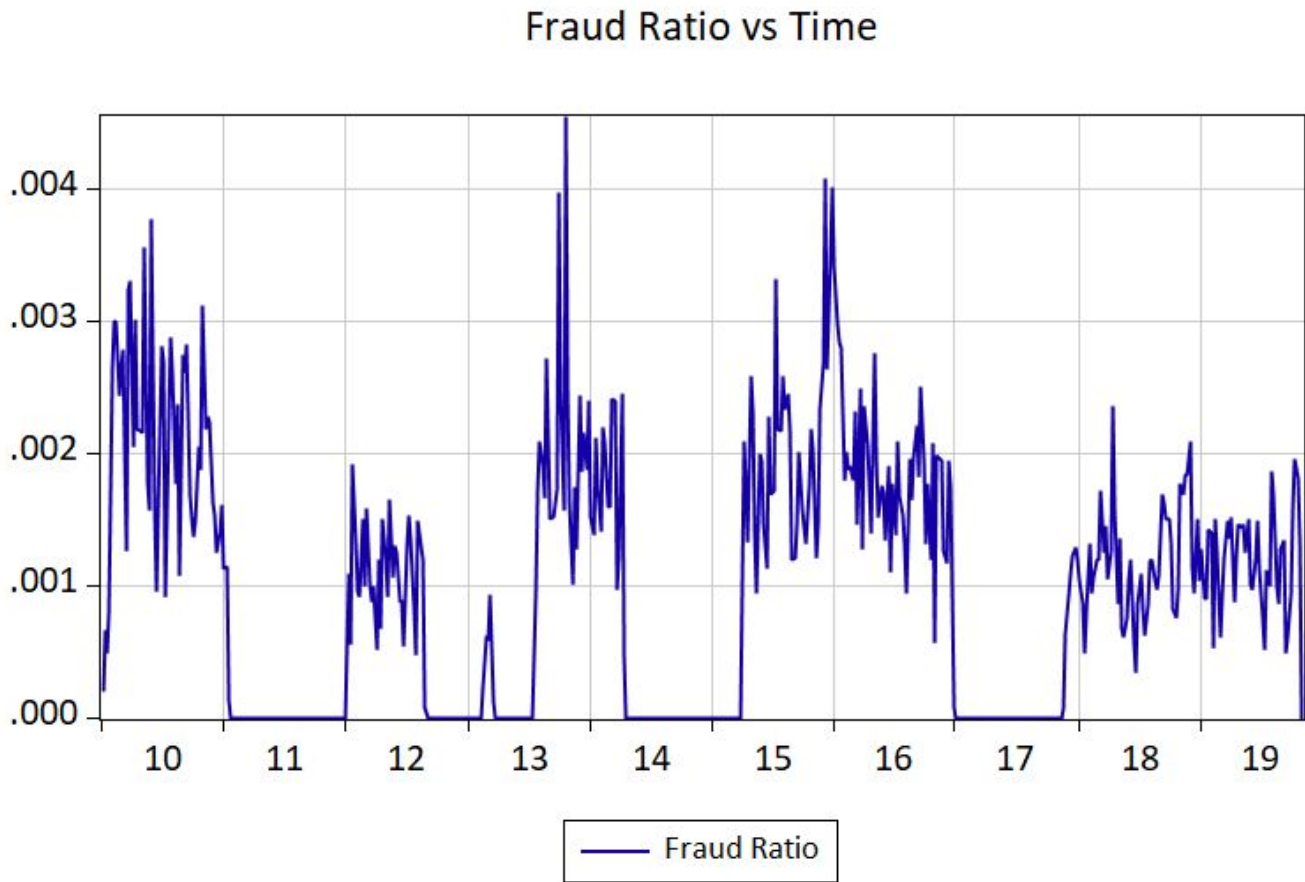
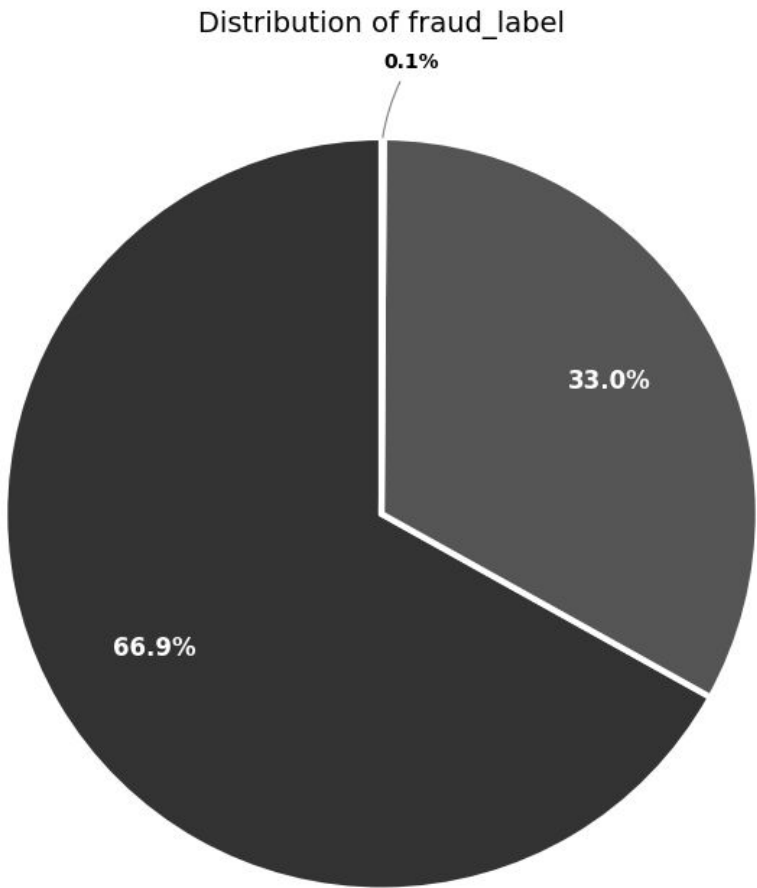
Errors	Frequency
No Error	13,094,522 (98.41%)
Insufficient Balance	130,902 (0.98%)
Bad Pin	32,119 (0.24%)
Technical Glitch	26,271 (0.20%)



Card on Dark Web	Frequency
No (0)	13,305,915 (100%)
Yes (1)	0 (0%)



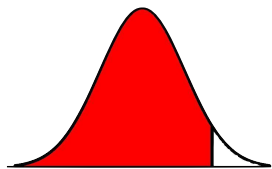
Categorical Variables—Fraud Label



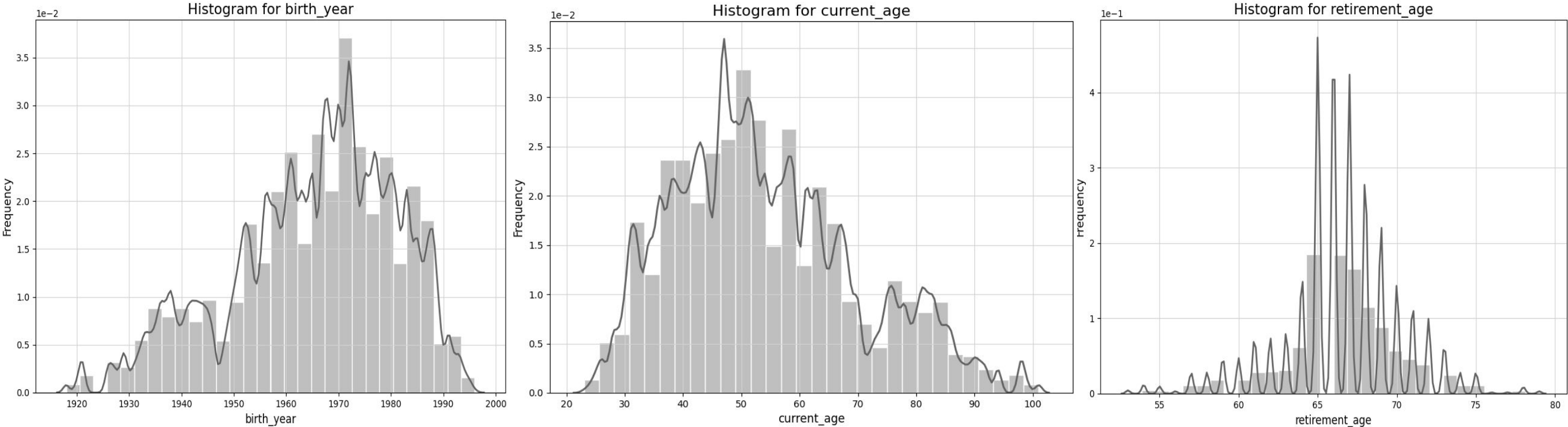
fraud_label
— No — Missing — Other

Fraud Label	Frequency
No	8,901,631 (66.90%)
Missing	4,390,952 (33.00%)
yes	13,332 (0.10%)

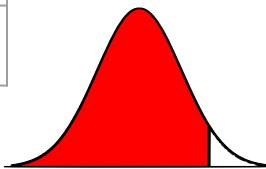
Fraud w/o Missing	Frequency
No	8,901,631 (99.85%)
Yes	13,332 (0.15%)



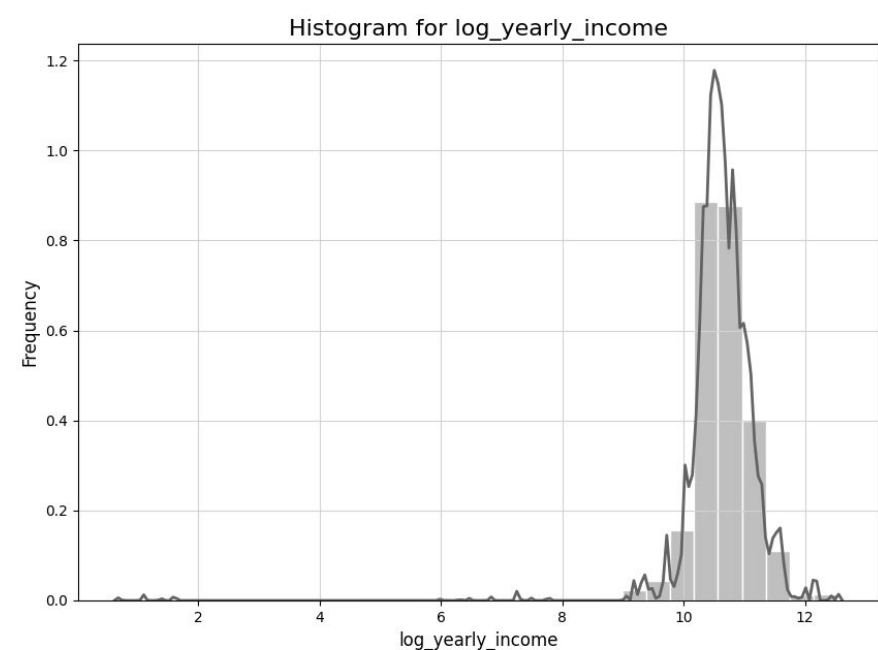
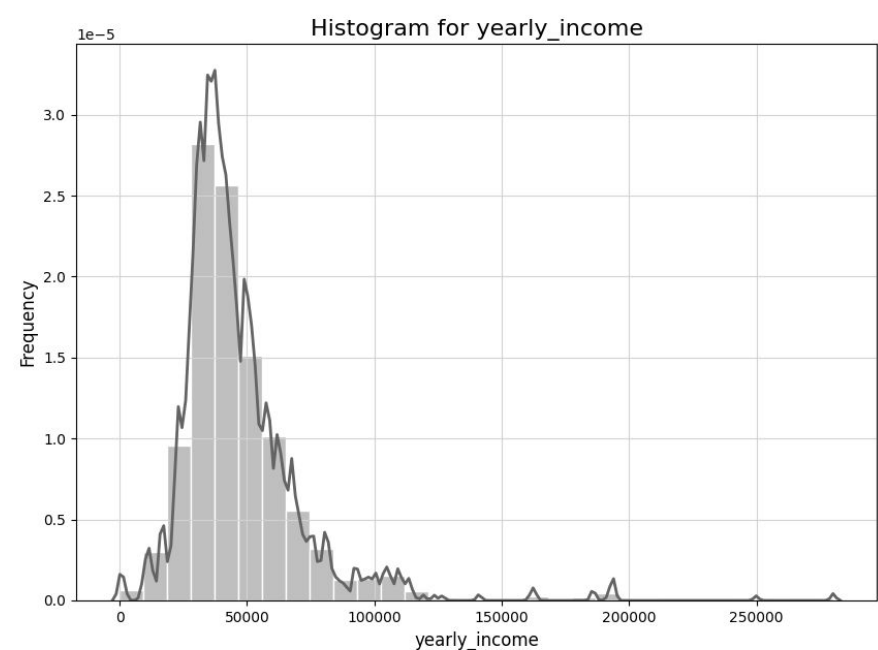
Continuous Variables—Client Age



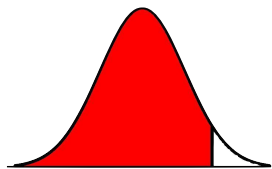
Statistics	birth_year	current_age	retirement_age
count	13,305,915	13,305,915	13,305,915
mean	1965.16	54.03	66.49
std	15.72	15.73	3.59
min	1918	23	53
25%	1956	42	65
50%	1968	52	66
75%	1977	63	68
max	1996	101	79
skewness	-0.55	0.55	-0.32
kurtosis	-0.27	-0.28	1.38



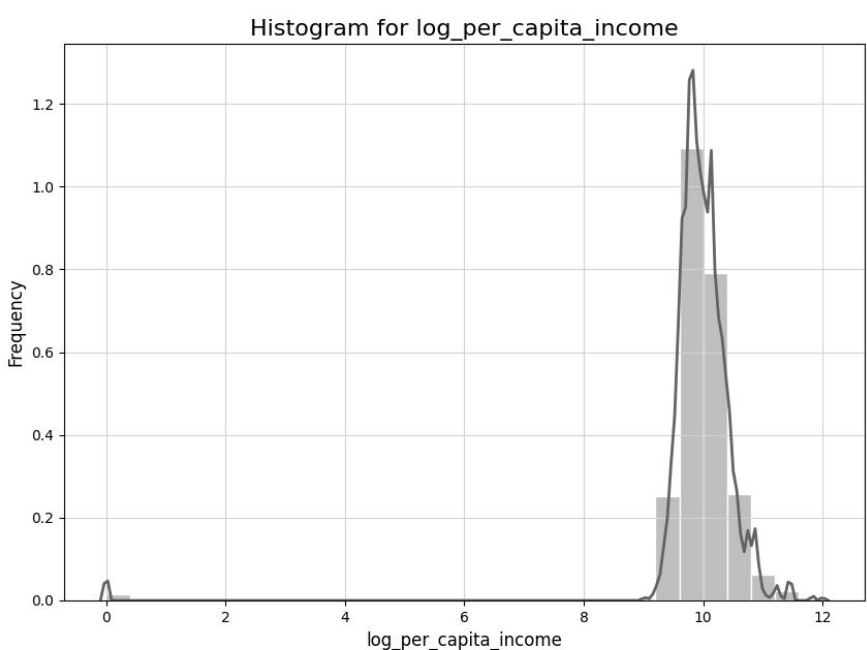
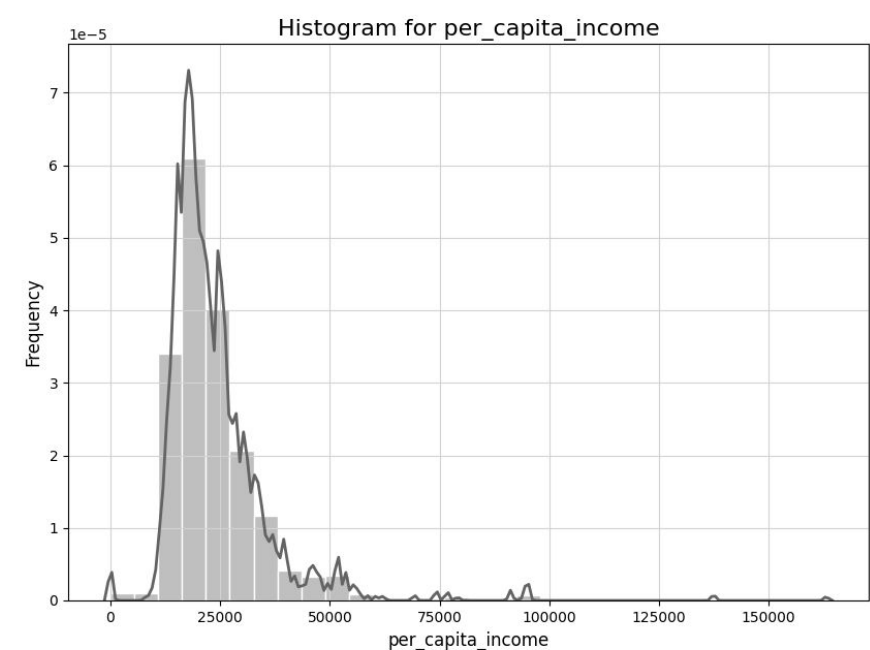
Continuous Variables—Income



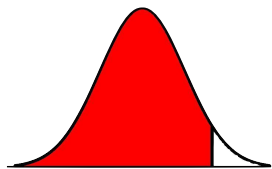
Statistics	yearly_income	log_yearly_income
count	13,305,915	13,305,915
mean	46,683.69	10.63
std	24,445.09	0.65
min	1	0.69
25%	32,817	10.40
50%	41,069	10.62
75%	54,013	10.90
max	280,199	12.54
skewness	3.11	-7.16
kurtosis	18.07	98.51



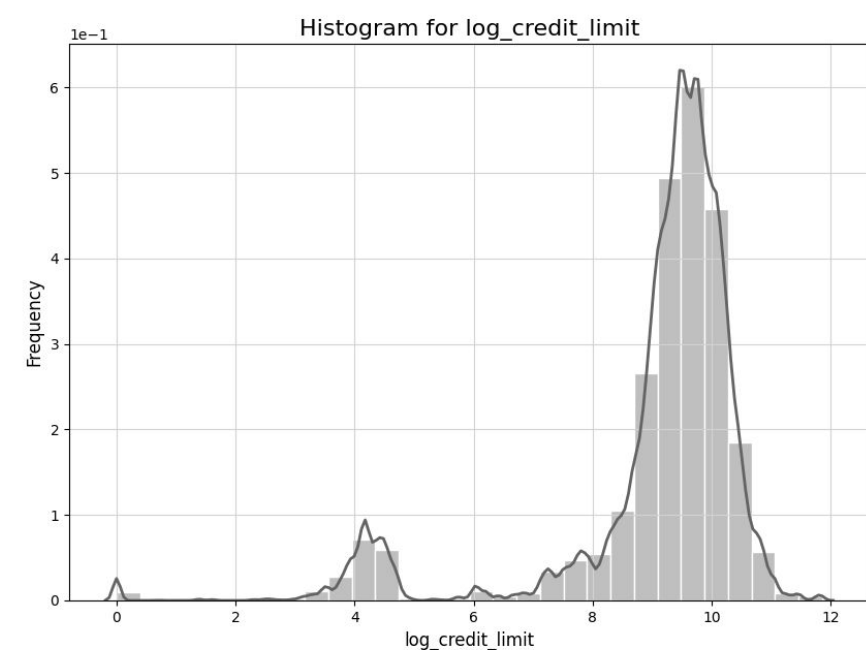
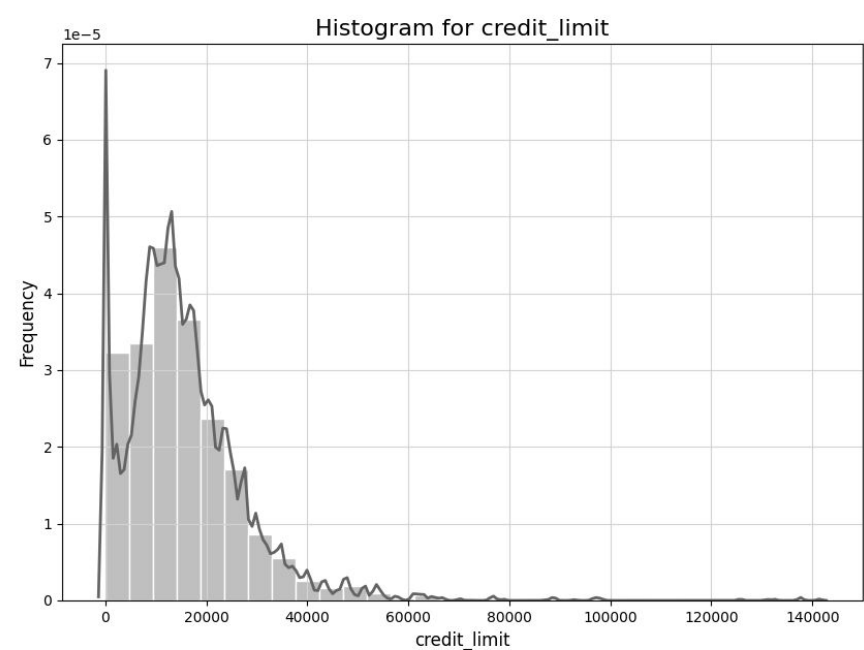
Continuous Variables—Income



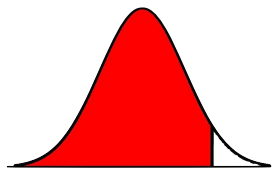
Statistics	per_capita_income	log_per_capita_income
count	13,305,915	13,305,915
mean	23,979.65	9.96
std	11,967.12	0.83
min	0	0
25%	17,113	9.75
50%	21,156	9.96
75%	27,308	10.21
max	163,145	12.00
skewness	3.72	-9.38
kurtosis	26.21	110.55



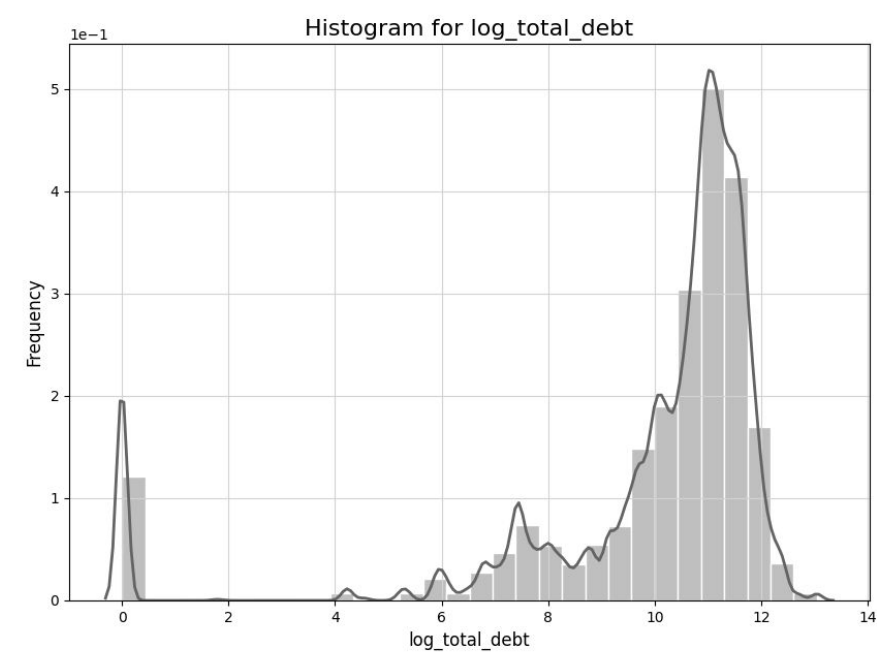
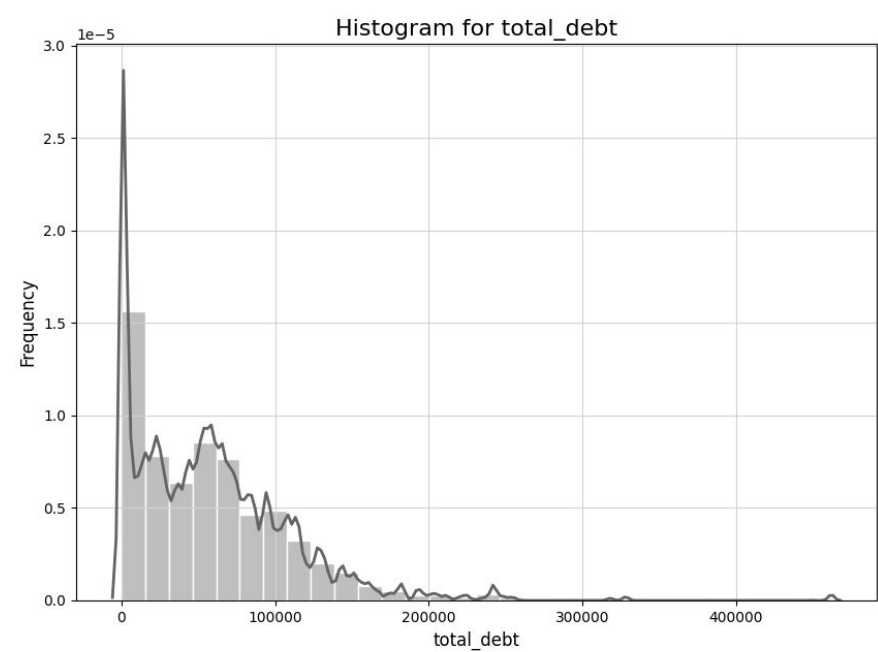
Continuous Variables—Credit Limit



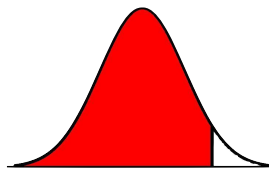
Statistics	credit_limit	log_credit_limit
count	13,305,915	13,305,915
mean	15,549.83	9.08
std	12,181.06	1.66
min	0	0
25%	8,100	9.00
50%	13,455	9.51
75%	20,839	9.94
max	141,391	11.86
skewness	2.42	-2.49
kurtosis	14.52	6.61



Continuous Variables—Debt

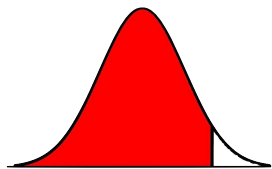
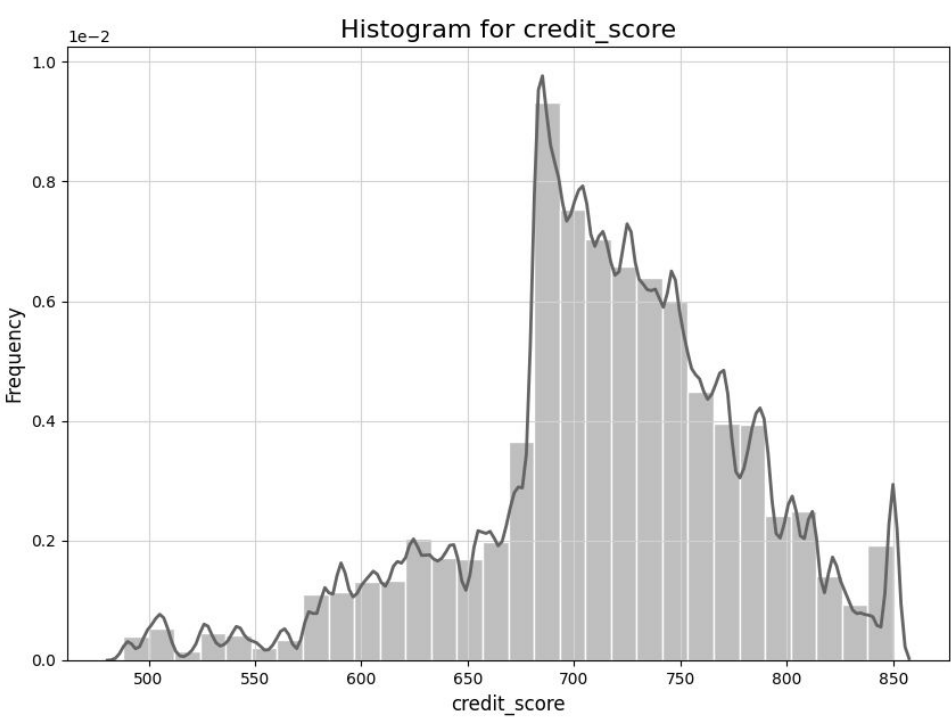


Statistics	total_debt	log_total_debt
count	13,305,915	13,305,915
mean	58,022.07	9.88
std	52,100.96	2.73
min	0	0
25%	16,437	9.71
50%	51,679	10.85
75%	85,160	11.35
max	461,854	13.04
skewness	1.74	-2.56
kurtosis	7.05	6.39

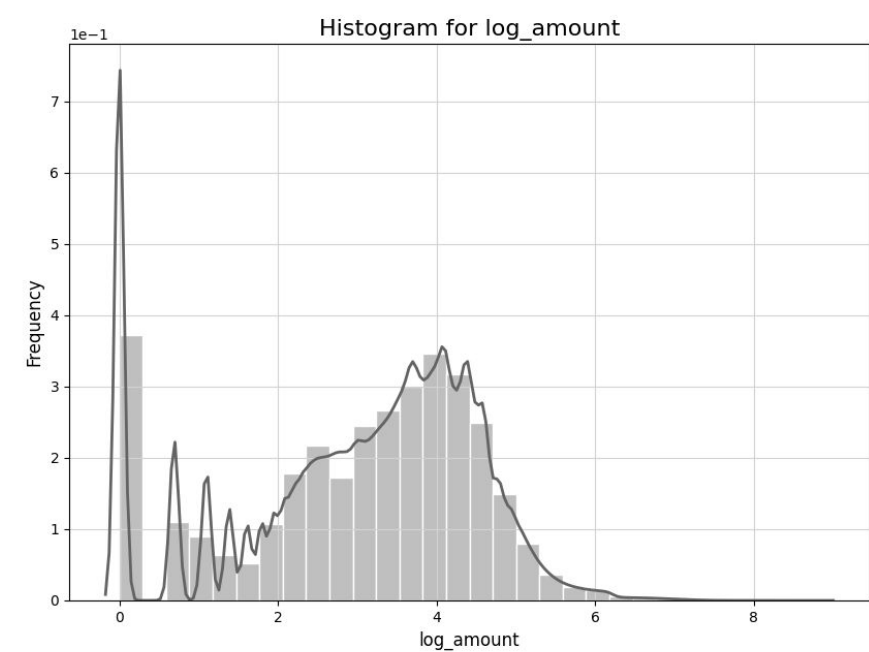
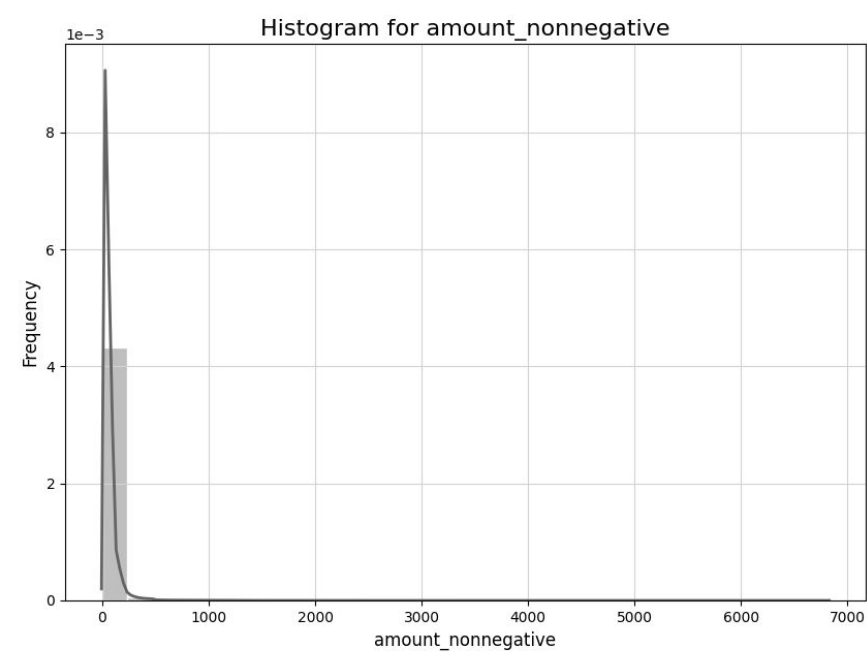


Continuous Variables—Credit Score

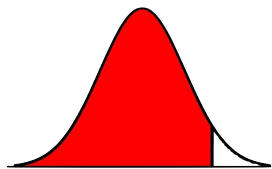
Statistics	credit_score
count	13,305,915
mean	713.91
std	65.80
min	488
25%	684
50%	716
75%	756
max	850
skewness	-0.60
kurtosis	0.82



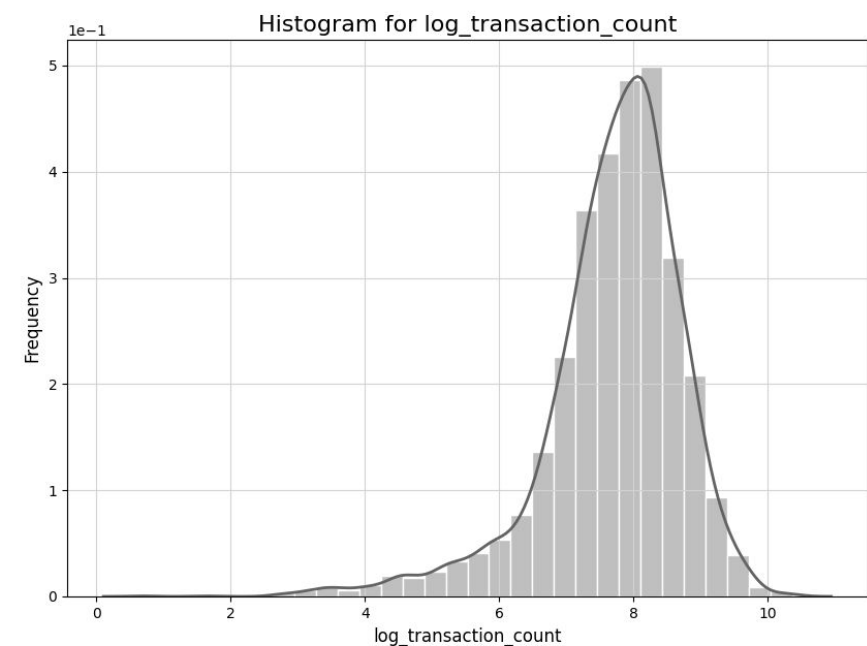
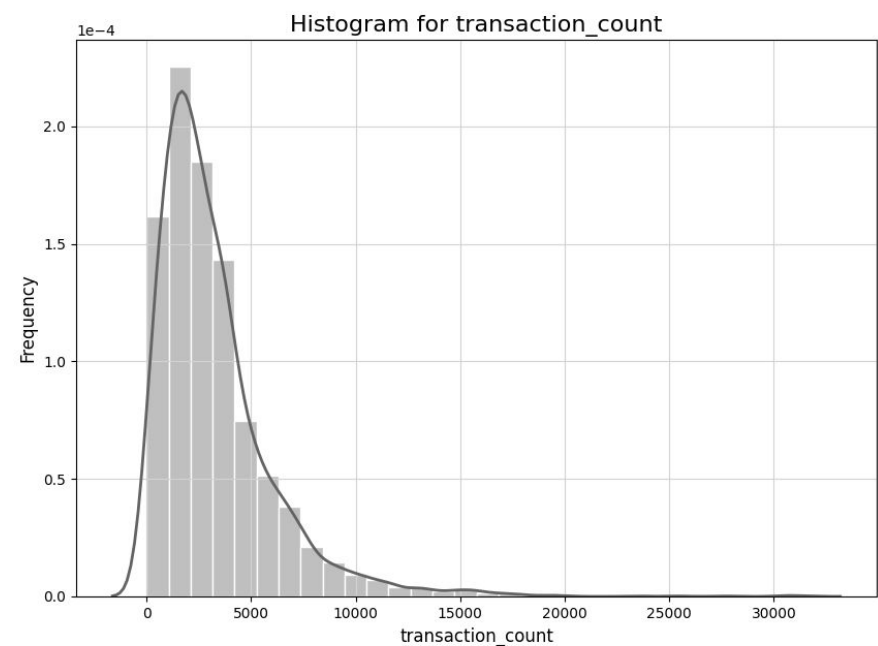
Continuous Variables—Amount



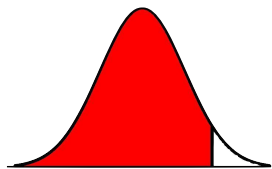
Statistics	amount/amount_nonnegative	log_amount*
count	13,305,915	13,305,915
mean	42.55/47.70	3.00
std	81.64/72.97	1.54
min	-500/1	0
25%	8	2.08
50%	28	3.33
75%	63	4.14
max	6,820	8.83
skewness	5.27/8.13	-0.58
kurtosis	111.28/168.48	-0.49



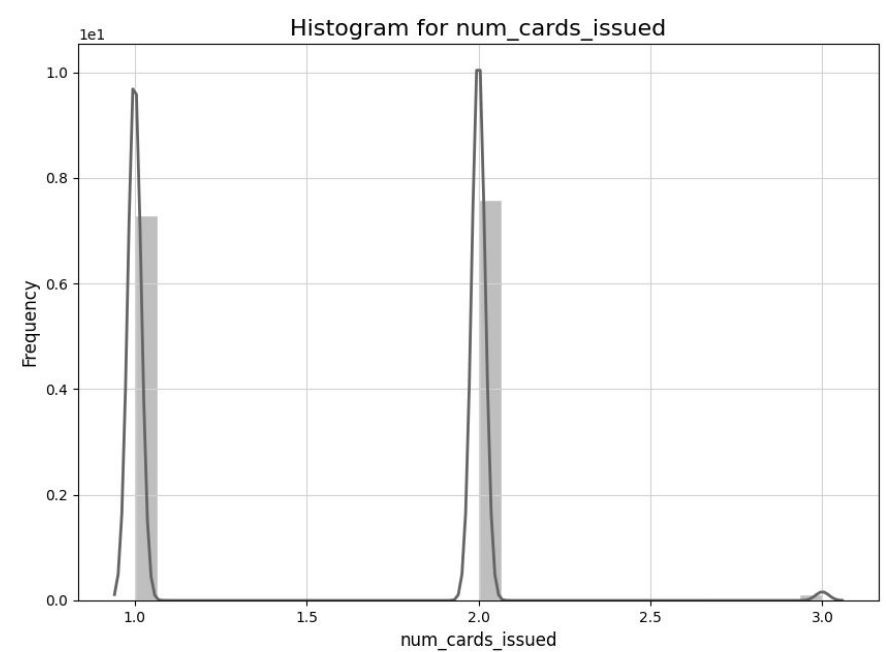
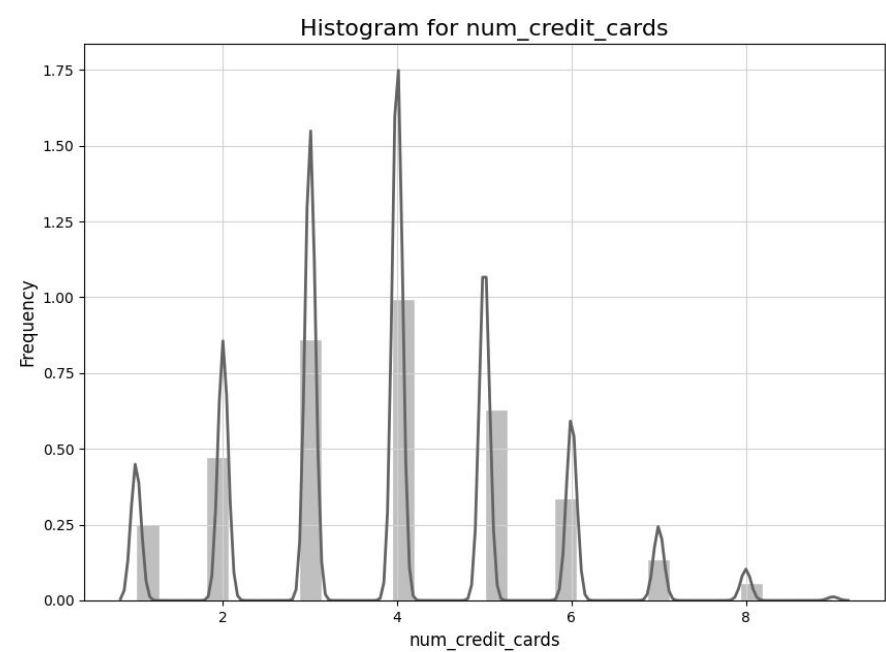
Continuous Variables—Transaction Frequency*



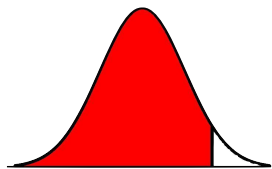
Statistics	transaction_count	log_transaction_count
count	4,071	4,071
mean	3,268.46	7.70
std	2,866.22	1.03
min	1	0.69
25%	1,419	7.26
50%	2,558	7.85
75%	4,166.5	8.34
max	31,552	10.36
skewness	2.58	-1.34
kurtosis	12.59	3.47



Continuous Variables—Number of Cards

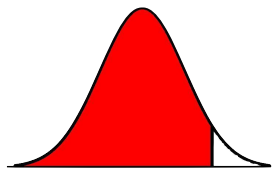
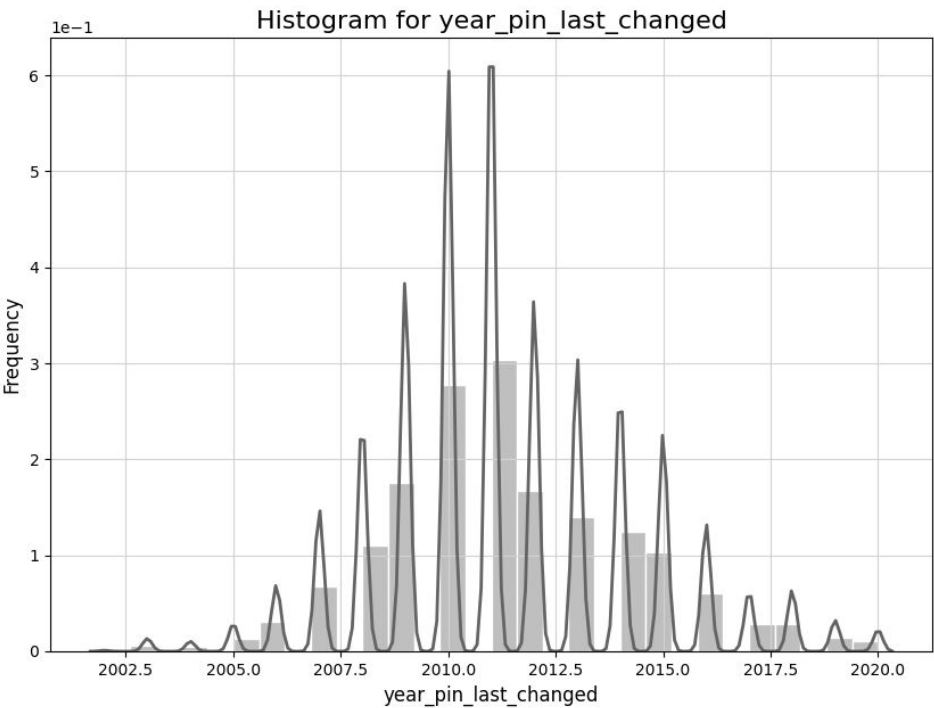


Statistics	num_credit_cards*	num_card_issued
count	13,305,915	13,305,915
mean	3.84	1.52
std	1.57	0.52
min	1	1
25%	3	1
50%	4	2
75%	5	2
max	9	3
skewness	0.31	0.08
kurtosis	-0.08	-1.57

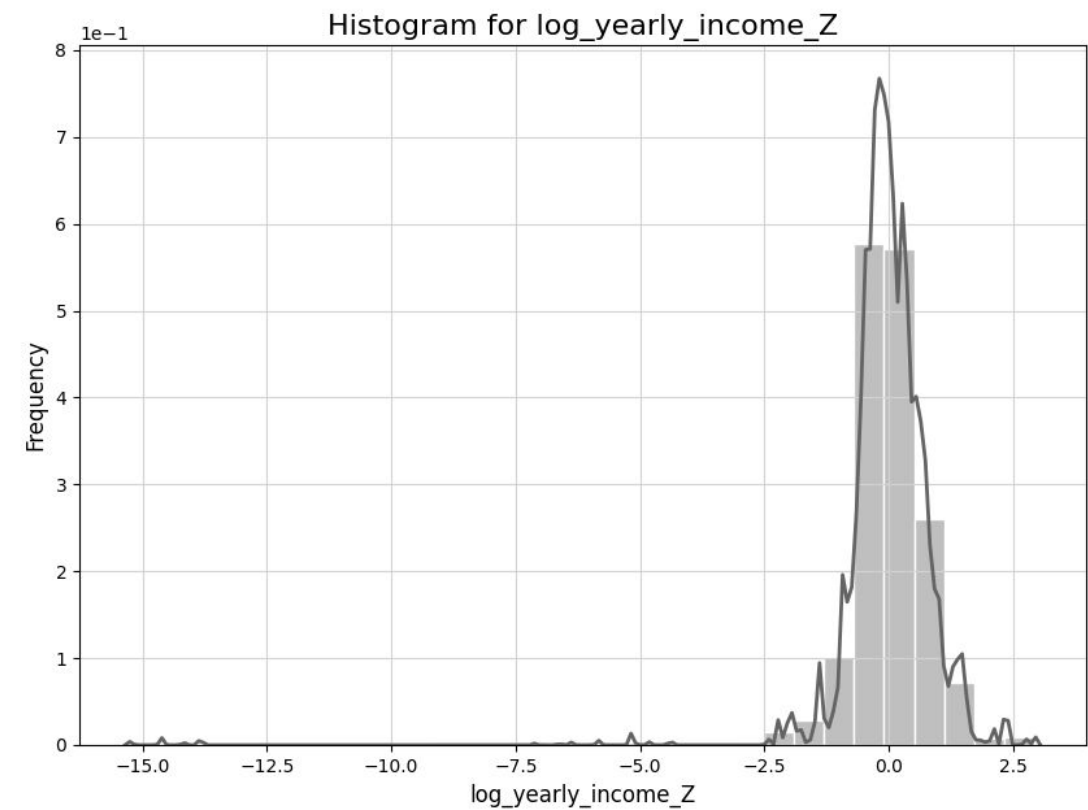
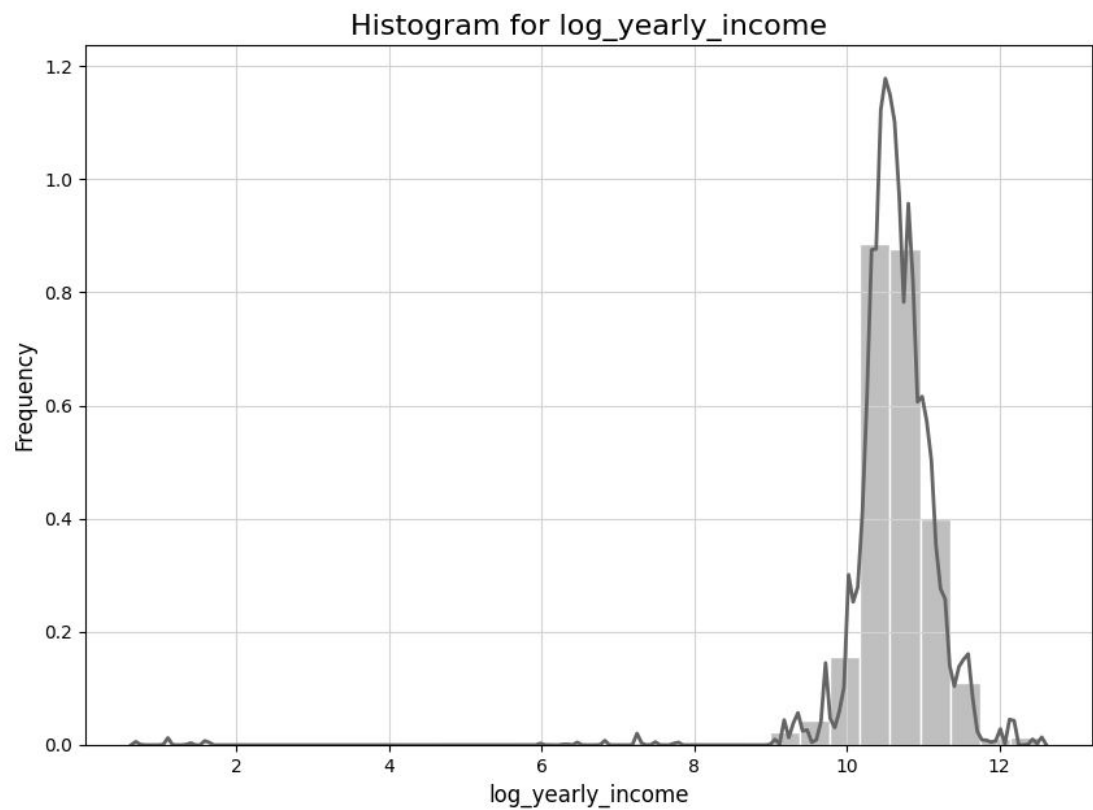


Continuous Variables—Year Pin

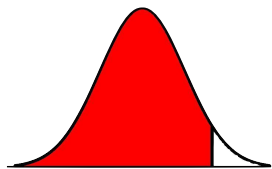
Statistics	year_pin_last_changed
count	13,305,915
mean	2,011.34
std	2.89
min	2002
25%	2010
50%	2011
75%	2013
max	2020
skewness	0.36
kurtosis	0.29



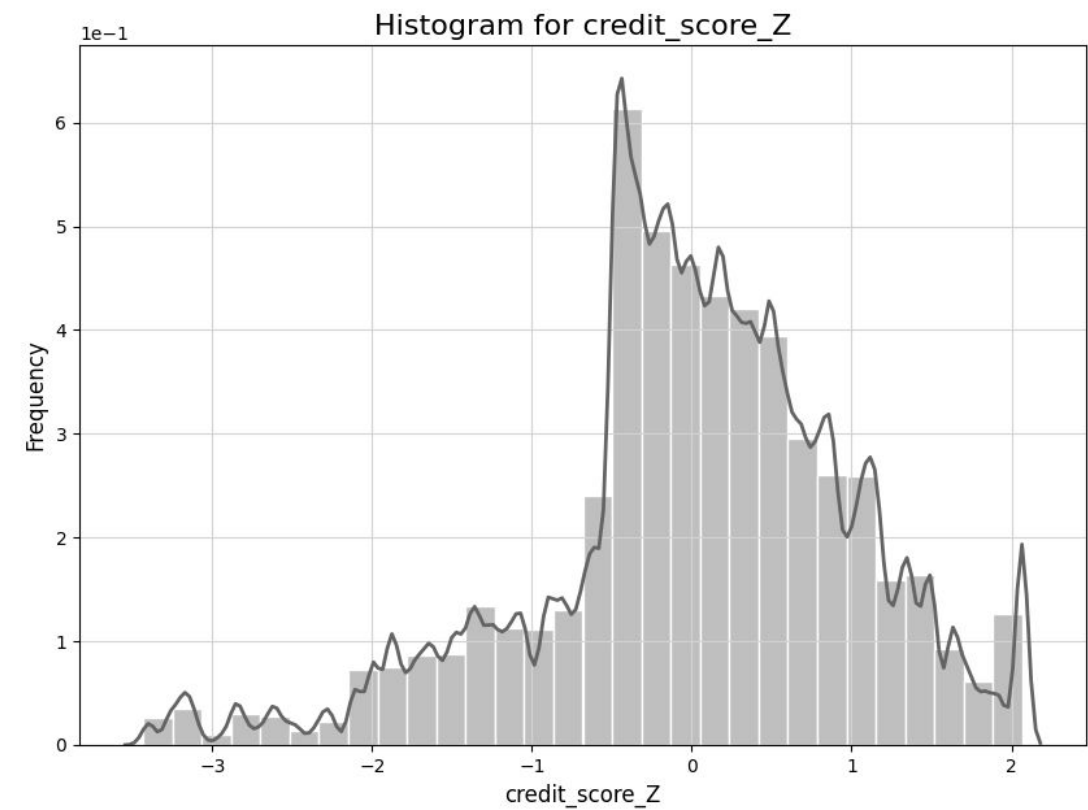
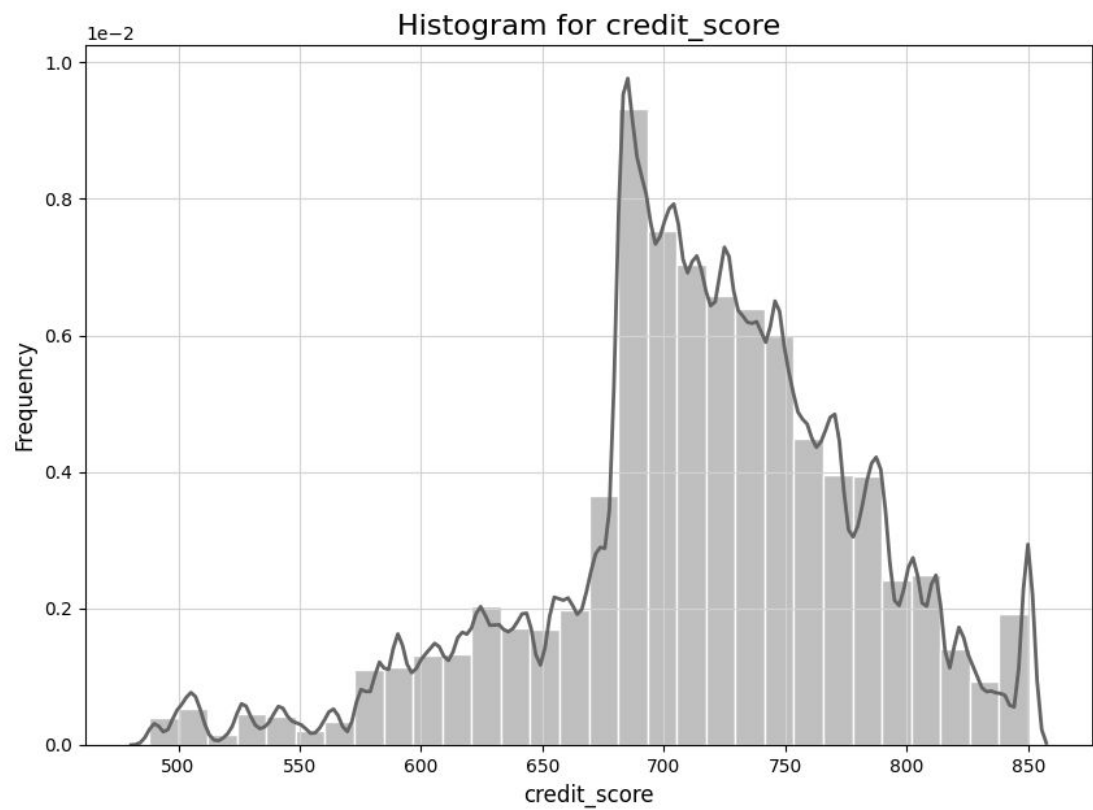
Scaling–Z score for Logarithm Variable



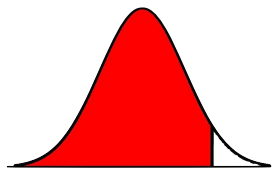
Original Variable	Logarithmic Variables	Scaled Variable	Reason
amount	log_amount	log_amount_Z	log transformation
yearly_income	log_yearly_income	log_yearly_income_Z	
per_capita_income	log_per_capita_income	log_per_capita_income_Z	
credit_limit	log_credit_limit	log_credit_limit_Z	
total_debt	log_total_debt	log_total_debt_Z	



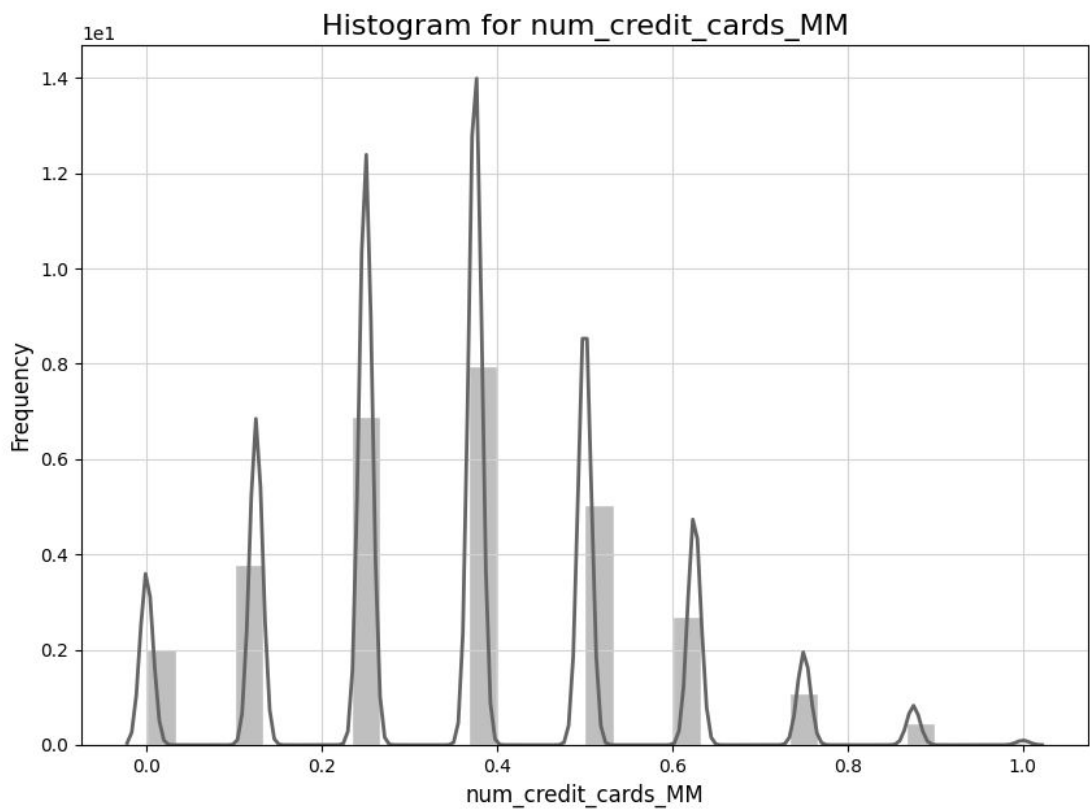
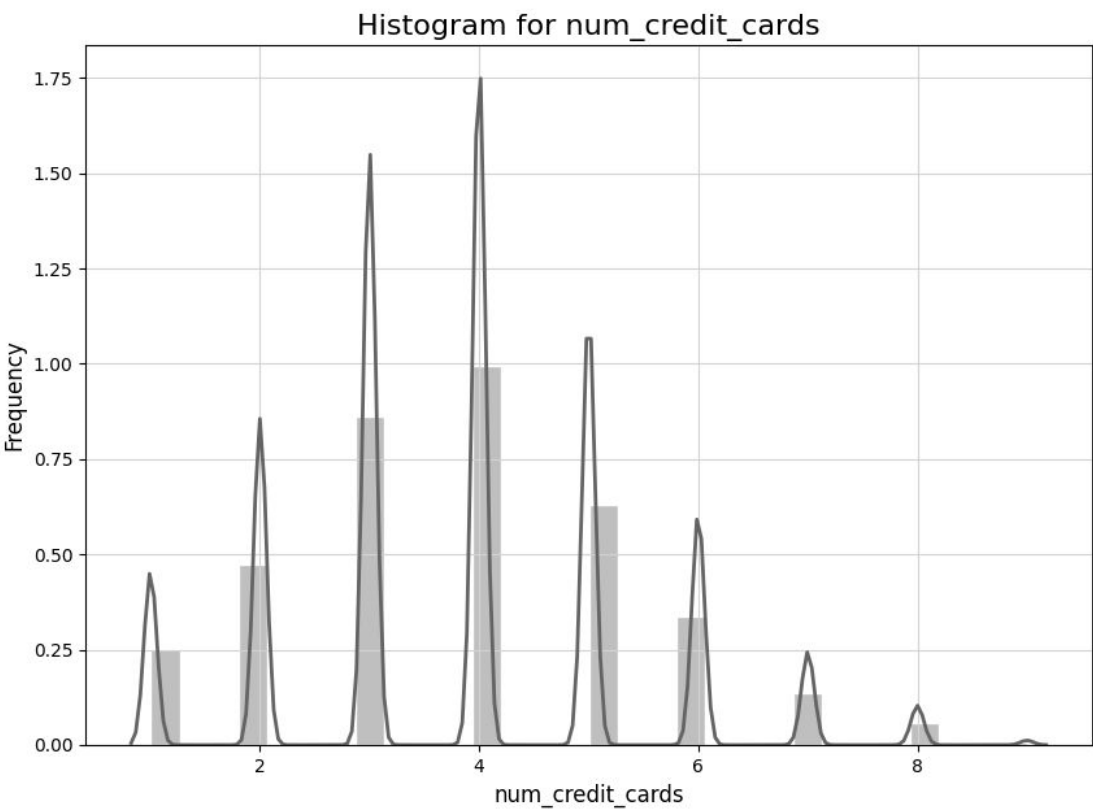
Scaling–Z score



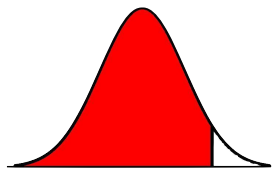
Variable	Scaled Variable	Reason
credit_score	credit_score_Z	highly bounded (Min 488.00, Max 850.00), close to normal, slightly left-skewed
current_age	current_age_Z	continuous, bounded variables, multimodal
retirement_age	retirement_age_Z	



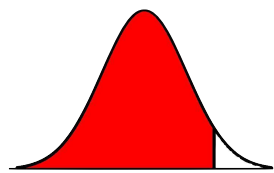
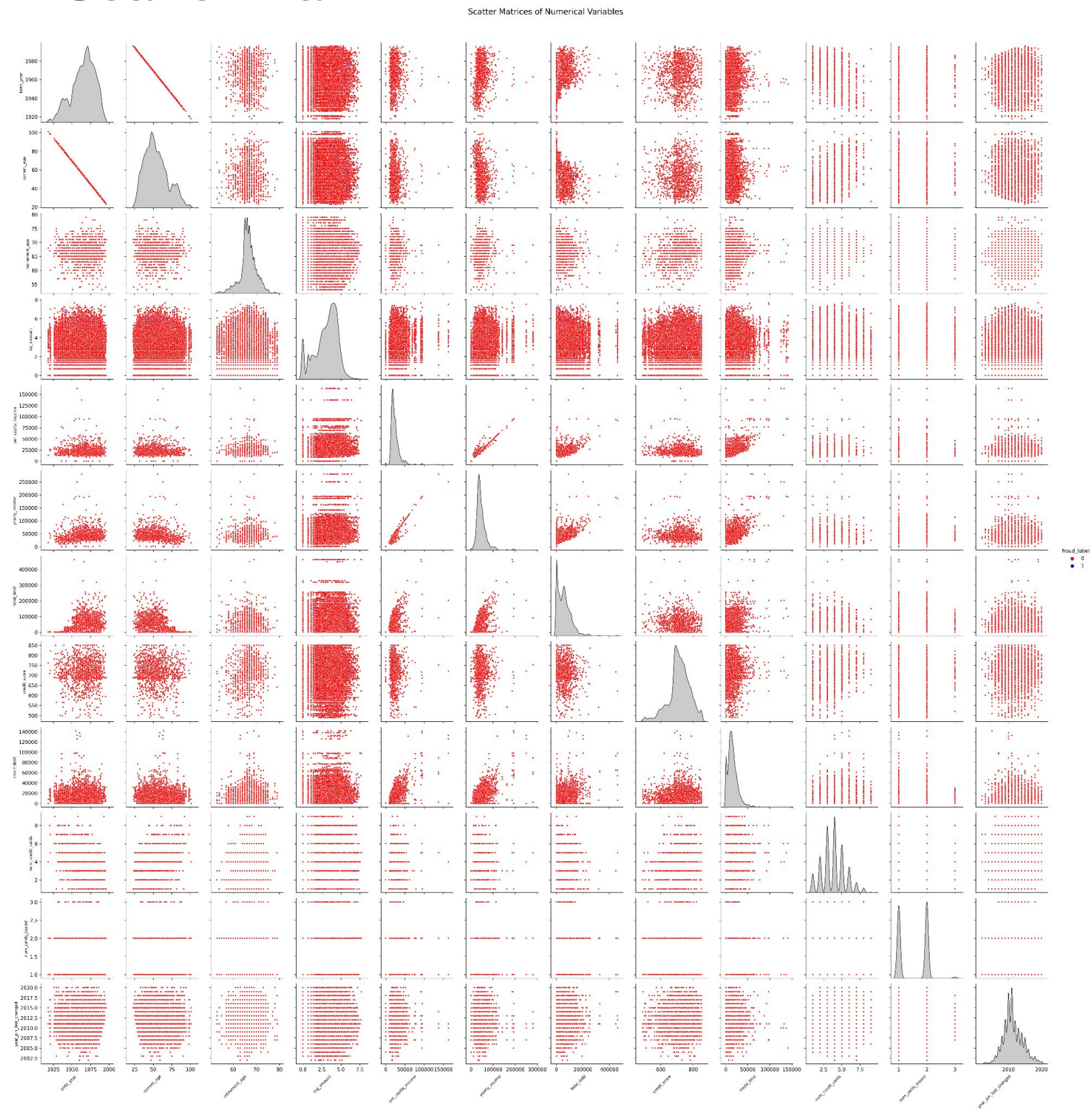
Scaling–Min-Max Scaling



Variable	Reason
num_credit_cards	ordinal, discrete, limited ranged (Min 1.00, Max 9.00)
num_cards_issued	extremely tightly bounded (Min 1.00, Max 3.00)



Correlation–Scatter Matrix

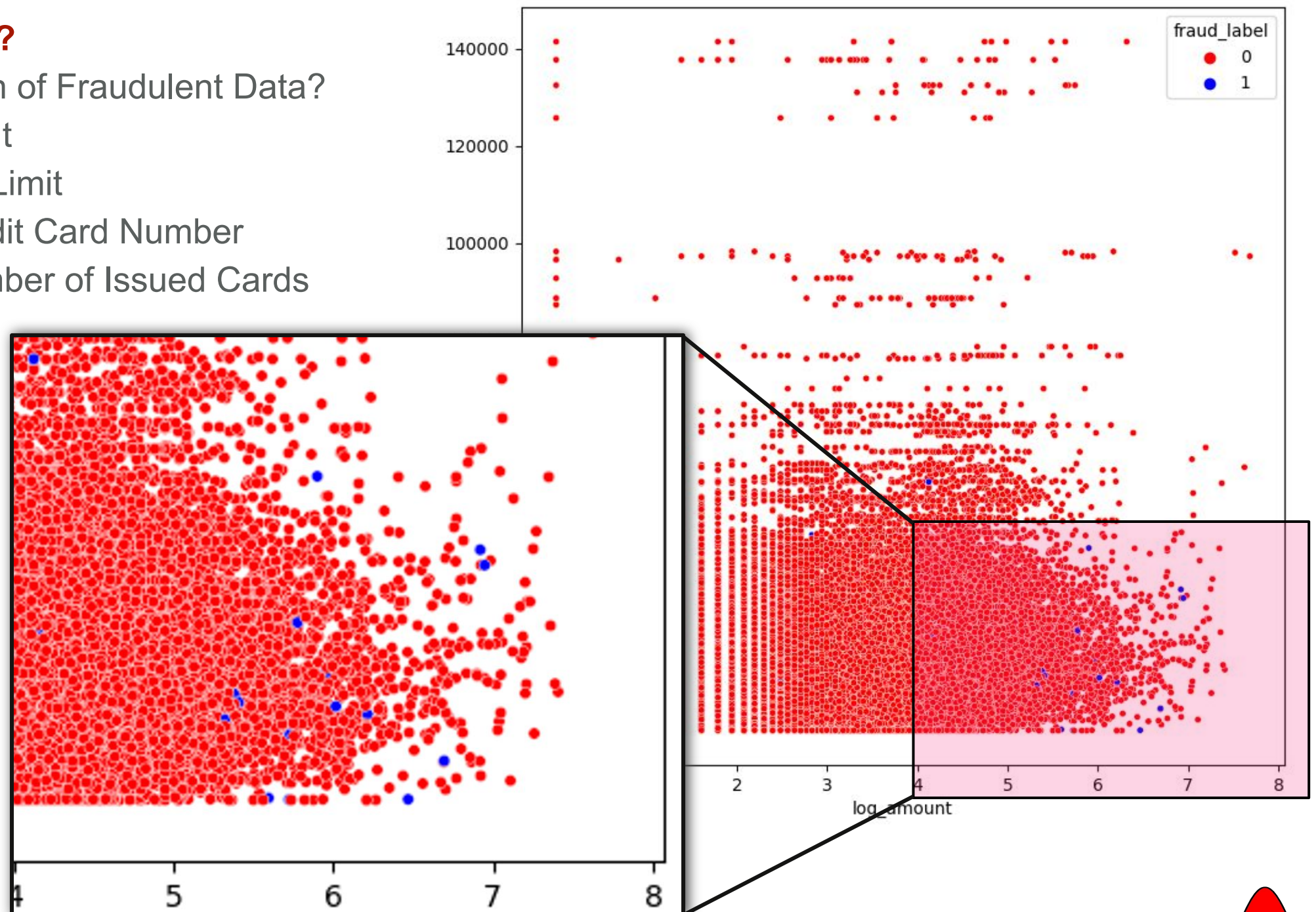


Correlation–Key Finding in Scatter Matrix

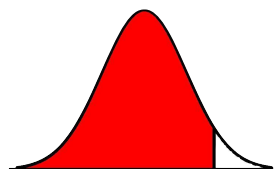
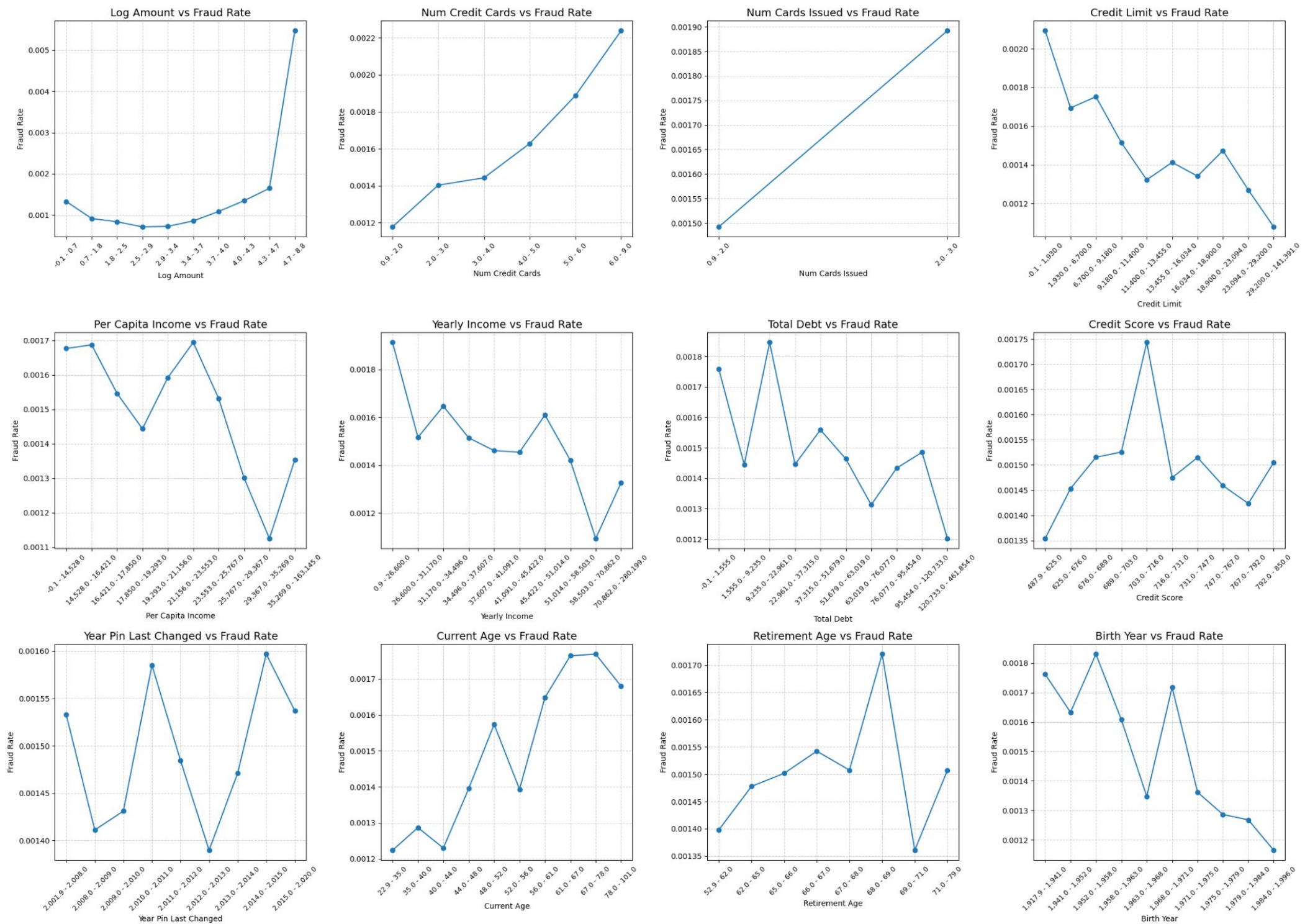
What's Special?

Potential Pattern of Fraudulent Data?

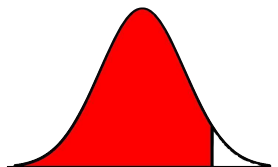
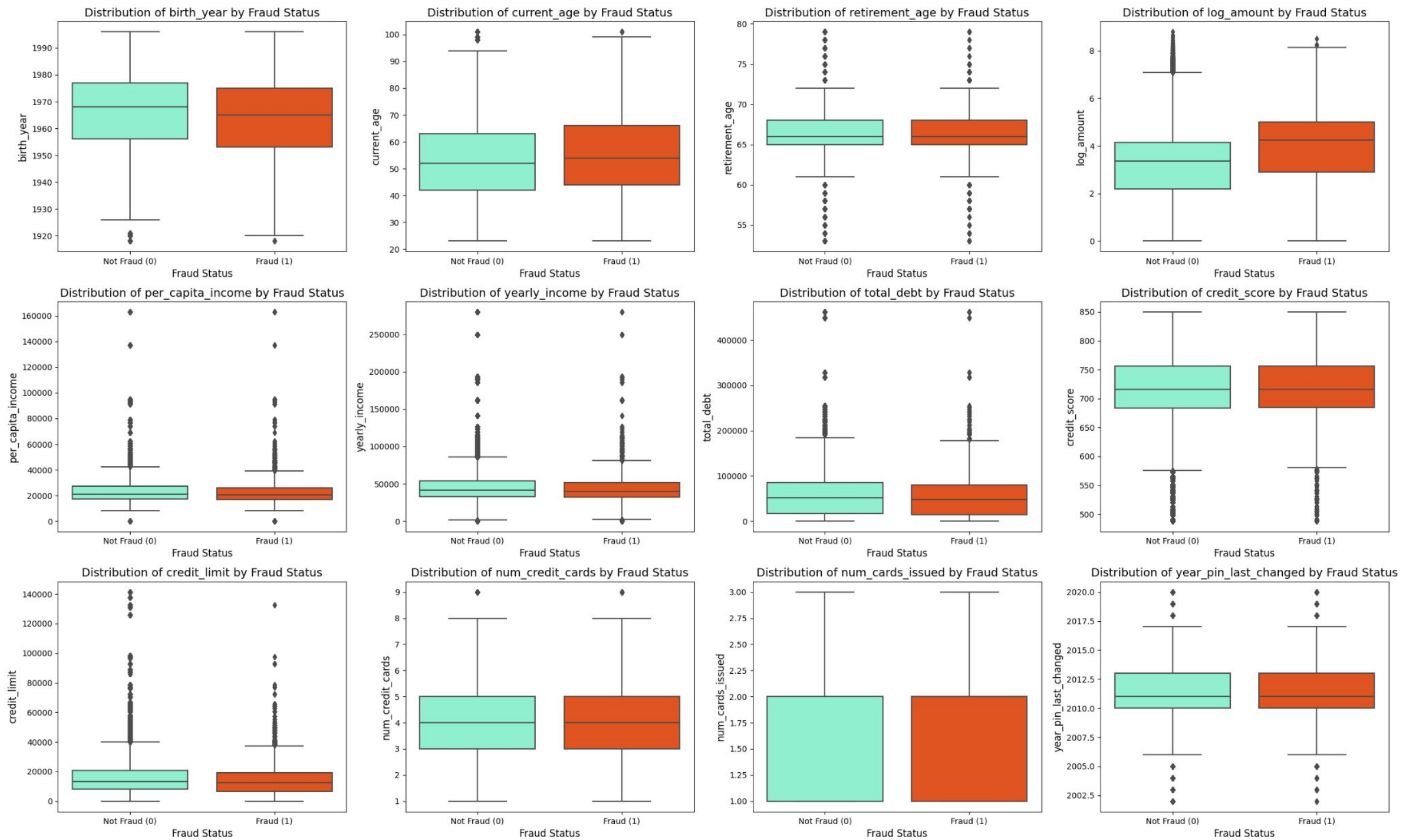
- High Amount
- Low Credit Limit
- High in Credit Card Number
- High in Number of Issued Cards



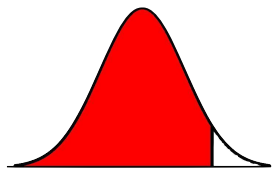
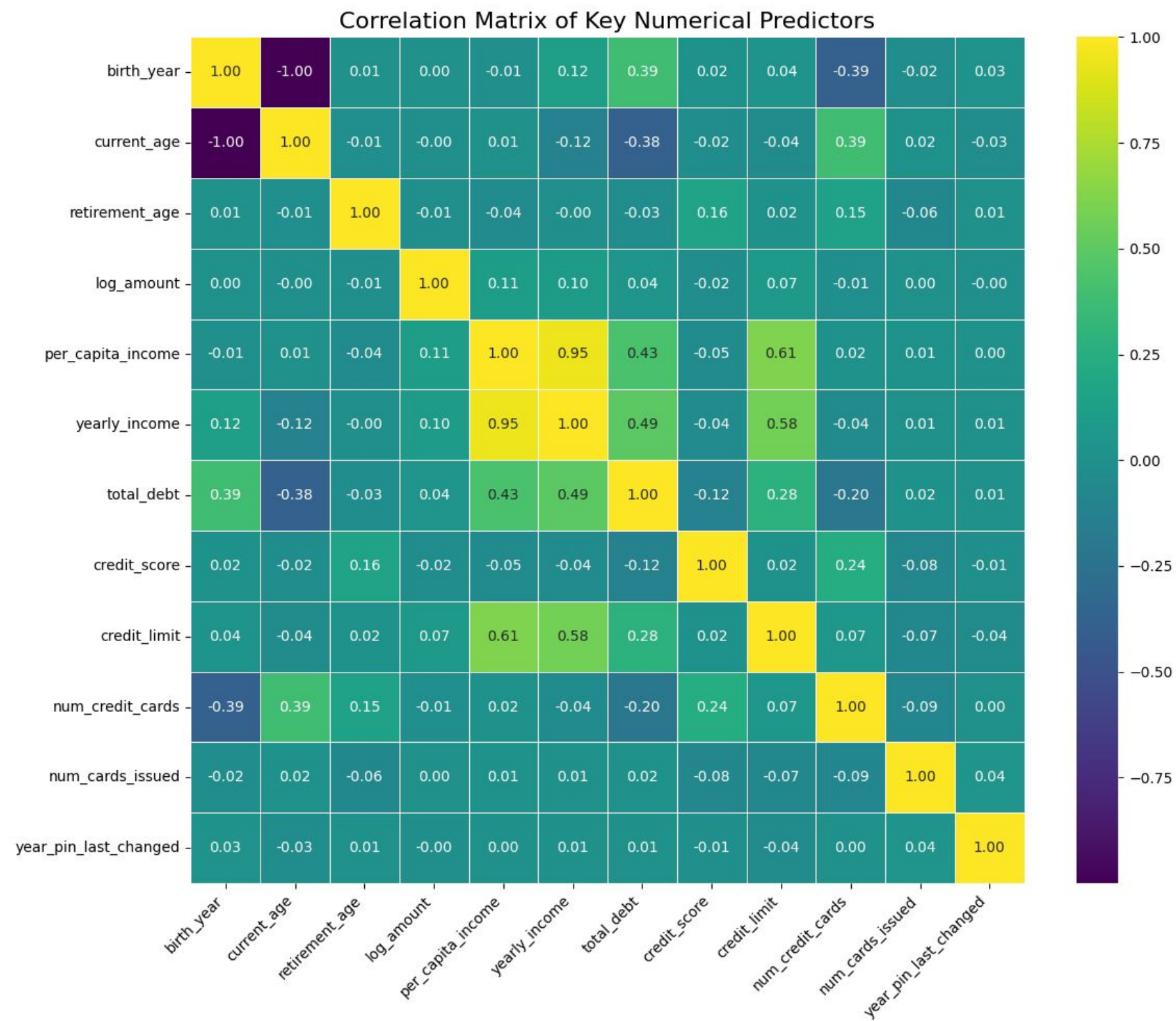
Correlation–Key Finding in Scatter Matrix



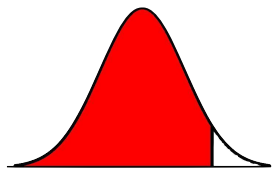
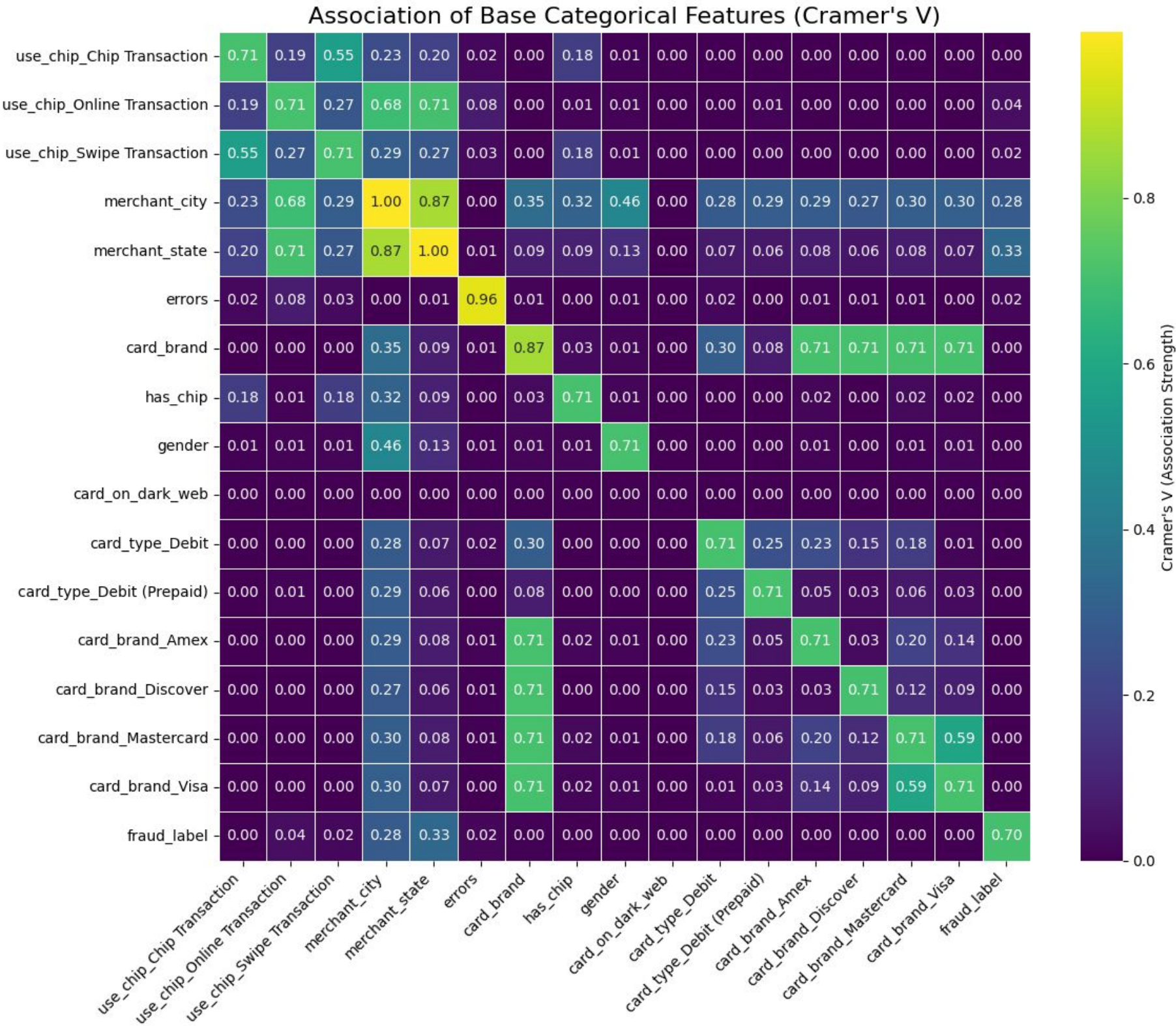
Correlation–Bivariate Box Plot



Correlation–Scatter Matrix

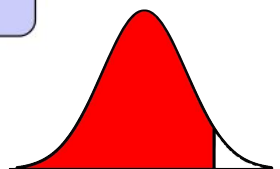
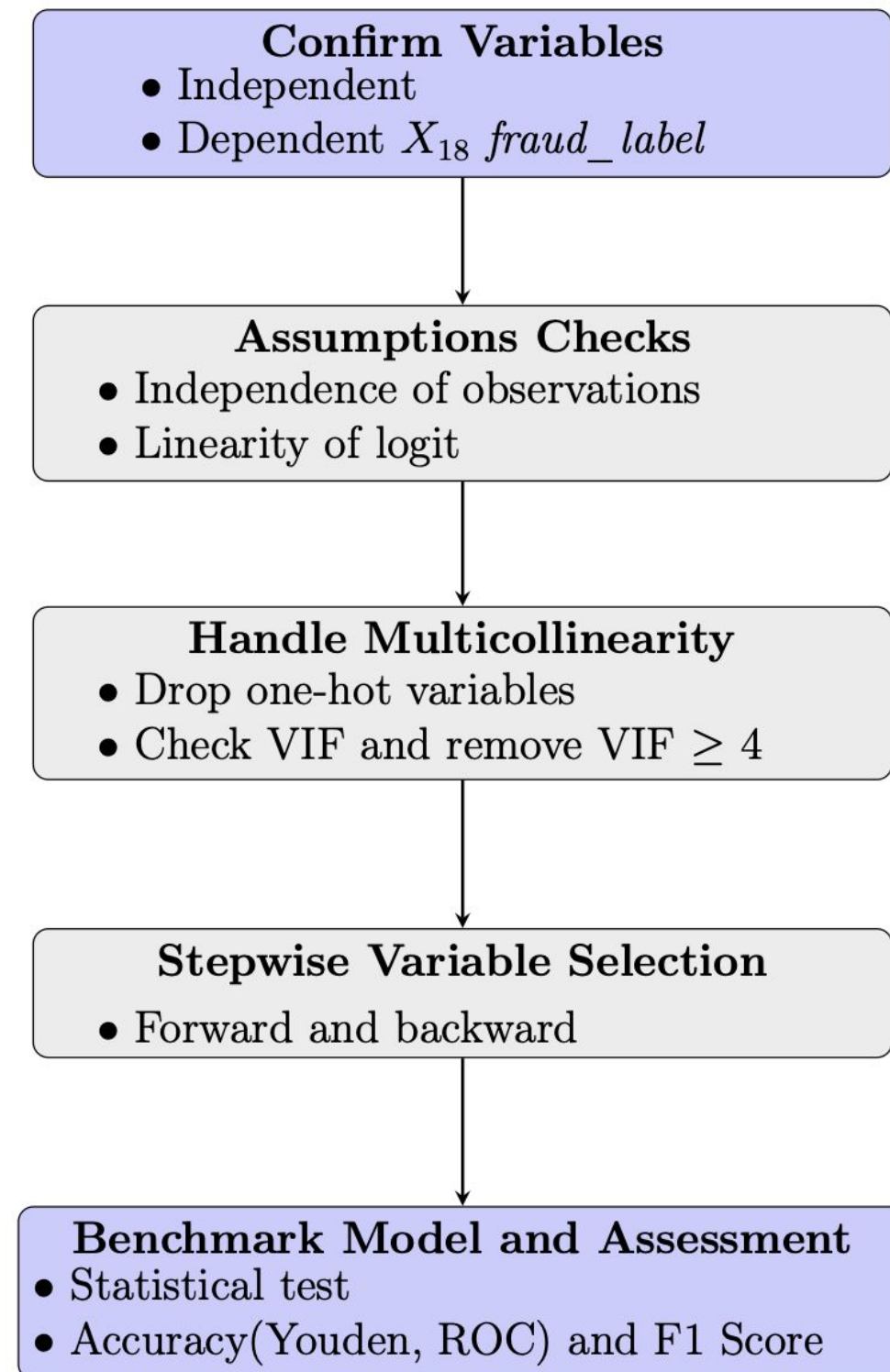


Correlation–Scatter Matrix



For benchmark model

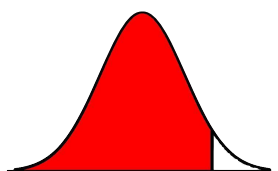
- **Aim:** handle binary classification issue
- **Modeling the Logistic Regression**
 - numeric/ categorical variables
 - assumptions
 - estimation
- **Logistic Regression Assessment**
- **Compare the Accuracy and F1 Score**



About Logistic Regression

- **Binary classification issue**
- **Few assumptions**
 - Independence of observations
 - Linearity of logit (Ignore)
- **Sample size (RULE of THUMB)**

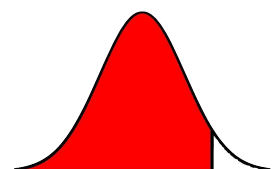
More than 400, 100 times of independents variables(29)
- **Training set:** 7,131,970
- **Testing set:** 1,782,994



Benchmark Model—Logistic Regression

Linearity of logit assumption– check Box Tidwell Test

Index	Variables	p-value	Result
X_5	amount	0.00E+00	Violated
X_{10}	zip	3.95E-197	Violated
X_{11}	mcc_code	4.86E-155	Violated
X_1	transaction_id	2.69E-49	Violated
X_7	merchant_id	7.49E-13	Violated
X_{20}	current_age	8.55E-11	Violated
X_{21}	retirement_age	3.01E-09	Violated
X_{35}	card_number	5.35E-05	Violated
X_{40}	credit_limit	5.78E-02	Satisfied
X_{31}	credit_score	6.75E-02	Satisfied
X_{32}	num_credit_cards	7.52E-02	Satisfied

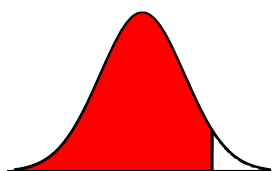


Benchmark Model—Logistic Regression

Multicollinearity assumption– check VIF index

Index	Variables	VIF Factor
X ₂₆	card_number	22.243090
X ₂₉	card_brand_Amex	4.419995
X ₃₈	zip	2.302851
X ₂₇	longitude	2.117485
X ₂₈	per_capita_income	1.843738
X ₁₆	use_chip_Chip Transaction	1.749374
X ₁₈	use_chip_Swipe Transaction	1.718199
X ₂₉	yearly_income	1.631605
X ₃₀	total_debt	1.538366
X ₁₄	zip_missing_flag	1.466621

Index	Variables	VIF Factor
X ₃₈	zip	2.53292
X ₂₇	longitude	2.460976
X ₁₈	use_chip_Swipe Transaction	1.979256
X ₁	transaction_id	1.768991
X ₄₀	credit_limit	1.707588
X ₂₉	yearly_income	1.627438
X ₃₀	total_debt	1.592191
X ₂₀	current_age	1.515529
X ₄₅	card_type_Debit	1.414488
X ₃₉	num_credit_cards	1.322601



Variables Selection guideline

1. Model Fit Evaluation:

Use the **-2 Log(Likelihood)** value to assess overall model improvement.

2. Stepwise Selection Procedure:

Iteratively add or remove variables based on statistical criteria.

3. Entry and Removal Criteria:

Include variables with **significant**($p < 0.05$) **LR test(add)/Wald statistic(remove)** ;
remove non-significant ones.

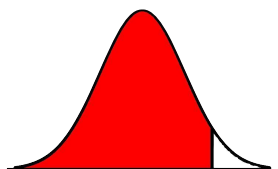
4. Stopping Rule:

Continue until **no excluded variable** produces a statistically significant improvement in model likelihood.

5. Goal:

Achieve a **parsimonious model** — good fit with minimal predictors.

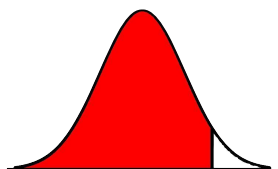
Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019).
Multivariate Data Analysis (8th ed.). Cengage Learning.



Variables for Logistic Regression

Summary of variables

Index	Variables	Type	Index	Variables	Type
Y	Fraud_label	numeric	X ₂₁	retirement_age	numeric
X ₁	transaction_id	numeric	X ₂₆	latitude	numeric
X ₃	client_id	numeric	X ₂₇	longitude	numeric
X ₅	amount	numeric	X ₂₉	yearly_income	numeric
X ₇	merchant_id	numeric	X ₃₁	credit_score	numeric
X ₁₀	zip	numeric	X ₃₂	num_credit_cards	numeric
X ₁₁	mcc_code	numeric	X ₃₈	has_chip	category
X ₁₅	errors_missing_flag	category	X ₄₀	credit_limit	numeric
X ₁₈	use_chip_Swipe Transaction	category	X ₄₉	card_brand_Masterc ard	category



Benchmark Model—Logistic Regression overall model fit

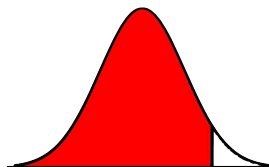
Stepwise completed until variables X_{49} card_brand_Mastercard

$$\begin{aligned} \log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = & -4.6829 + 0.3586X_5 - 2.1209X_{18} - 7.17 \times 10^{-8}X_1 \\ & - 6.43 \times 10^{-5}X_{10} - 2.6983 \times 10^{-2}X_{27} - 4.2024 \times 10^{-3}X_{40} \\ & + 9.74 \times 10^{-6}X_7 + 0.1061X_{32} - 0.7411X_{15} \\ & - 0.0297X_{29} - 1.50 \times 10^{-4}X_{11} - 0.3366X_{38} \\ & - 0.0167X_{26} - 1.16 \times 10^{-4}X_3 + 7.44 \times 10^{-4}X_{31} \\ & + 0.0108X_{21} + 0.0514X_{49} \end{aligned}$$

Train Accuracy: 0.9985, Test Accuracy: 0.9985

=== Overall Model Fit ===

		Measure	Value	Change_from_Base	Change_pvalue
0	-2	Log Likelihood (-2LL) value	131399.390	27711.222	0.0
1		Cox and Snell R2	0.004		
2		Nagelkerke R2	0.176		
3		Pseudo R2 (McFadden)	0.174		
4		Hosmer-Lemeshow χ^2	27711.222		

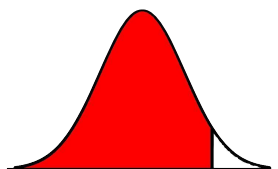


Benchmark Model—Logistic Regression overall model fit

Stepwise completed until variables X_{49} card_brand_Mastercard

===== Variables in the Equation =====

Index	Independent Variable	B	Std. Error	Wald	df	Sig.	Exp(B)
0	const	-4.682906	0.253341	341.679	1	2.75e-76	0.009252
1	amount	0.358623	0.008900	1623.813	1	0.000000	1.431356
2	use_chip_Swipe Transaction	-2.120852	0.035093	3652.318	1	0.000000	0.119929
3	transaction_id	-0.000000	0.000000	1000.211	1	1.62e-219	1.000000
4	zip	-0.000064	0.000001	5076.823	1	0.000000	0.999936
5	longitude	-0.026983	0.000566	2269.179	1	0.000000	0.973378
6	credit_limit	-0.004202	0.000215	383.313	1	2.36e-85	0.995806
7	merchant_id	0.000010	0.000000	695.384	1	3.02e-153	1.000010
8	num_credit_cards	0.106118	0.006416	273.587	1	1.88e-61	1.111953
9	errors_missing_flag	-0.741082	0.048343	234.997	1	4.85e-53	0.476598
10	yearly_income	-0.029681	0.002085	202.598	1	5.66e-46	0.970755
11	mcc_code	-0.000150	0.000012	163.726	1	1.74e-37	0.999850
12	has_chip_YES	-0.336632	0.035027	92.364	1	7.21e-22	0.714171
13	latitude	-0.016679	0.001947	73.396	1	1.06e-17	0.983459
14	client_id_x	-0.000116	0.000017	46.984	1	7.16e-12	0.999884
15	credit_score	0.000744	0.000158	22.115	1	2.57e-06	1.000745
16	retirement_age	0.010763	0.002892	13.848	1	1.98e-04	1.010821
17	total_debt	0.000294	0.000099	8.785	1	3.04e-03	1.000294
18	card_brand_Mastercard	0.051355	0.019647	6.832	1	8.95e-03	1.052696



Benchmark Model—Logistic Regression overall model fit

Stepwise completed until variables X_{49} card_brand_Mastercard

=== Variables Not in the Equation ===

	Independent Variable	Score Statistic (LRT)	Significance
0	card_id	2.973	0.0847
1	current_age	0.080	0.7767
2	gender_Male	0.071	0.7905
3	cvv	1.239	0.2658
4	num_cards_issued	0.003	0.9595
5	card_type_Debit	2.495	0.1142

=====

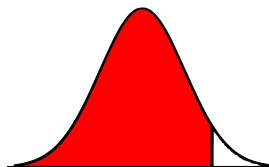
=== Accuracy in Training and Testing dataset ===

=== Classification Matrix – Training Sample ===

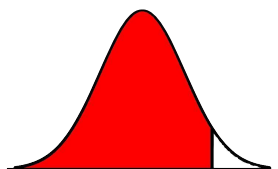
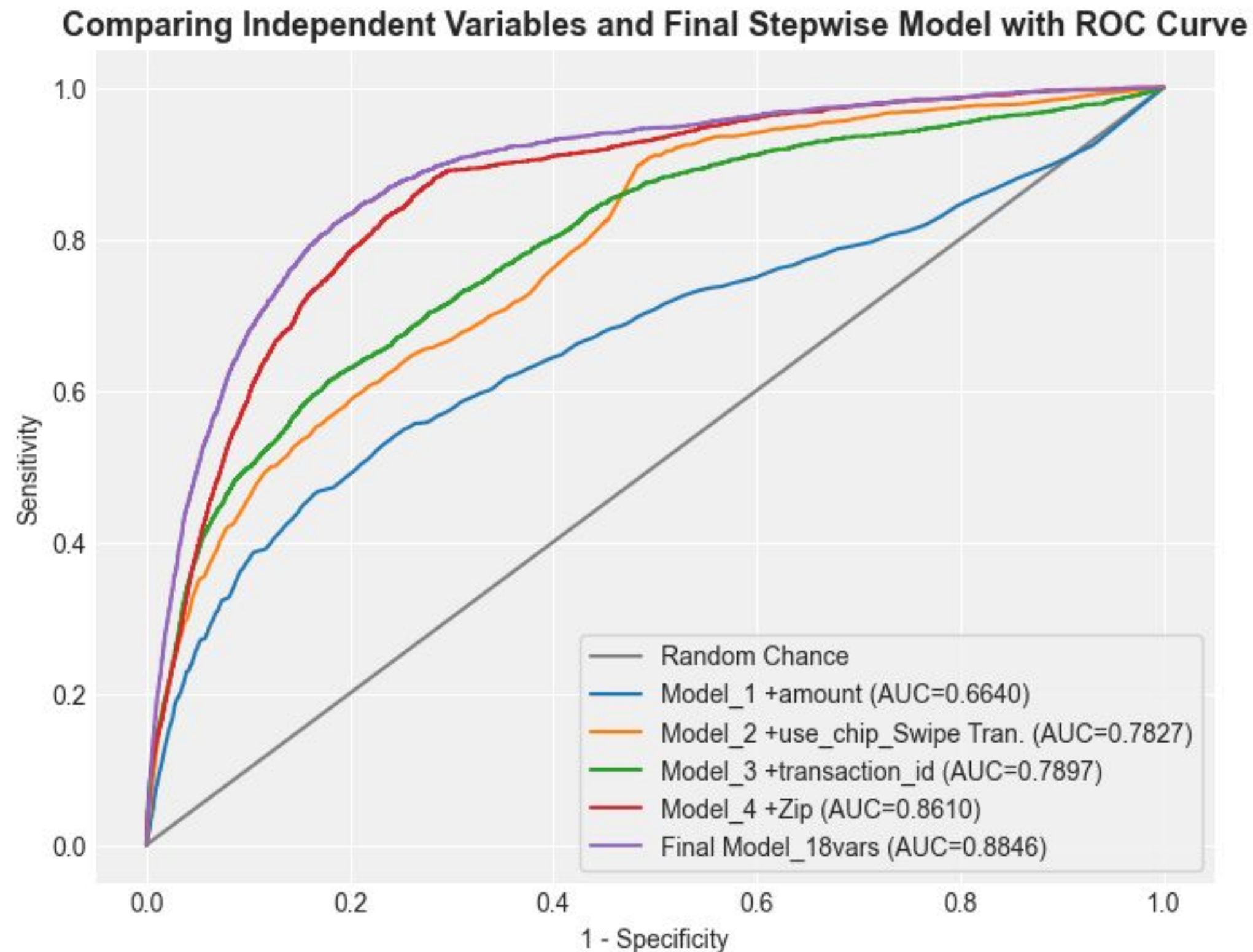
Actual Group	Predicted Fraud (1)	Predicted Normal (0)	Total	% Correct	cutoff	F1 Score
Fraud (1)	9177	1414	10591	86.6	0.0015	0.0114
Normal (0)	1583470	5537909	7121379	77.8		
Total	1592647	5539323	7131970	77.8		

=== Classification Matrix – Holdout (Test) Sample ===

Actual Group	Predicted Fraud (1)	Predicted Normal (0)	Total	% Correct	cutoff	F1 Score
Fraud (1)	2333	408	2741	85.1	0.0015	0.0116
Normal (0)	395986	1384266	1780252	77.8		
Total	398319	1384674	1782993	77.8		



Benchmark Model—Logistic Regression overall model fit

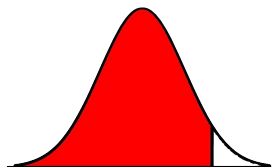


Benchmark Model—Logistic Regression overall model fit

Cutoff	TN	FP	FN	TP	Accuracy	Sensitivity	Specificity	Youden Index	PPV	NPV	F1 Score
0.0000	0	1780252	0	2741	0.20%	100.00%	0.00%	0.00%	0.20%	NC	0.0031
0.0005	1106443	673809	208	2533	62.20%	92.40%	62.20%	54.60%	0.40%	100.00%	0.0075
0.0010	1290076	490176	303	2438	72.50%	88.90%	72.50%	61.40%	0.50%	100.00%	0.0098
0.0015	1384266	395986	408	2333	77.80%	85.10%	77.80%	62.90%	0.60%	100.00%	0.0116
0.0020	1449735	330517	494	2247	81.40%	82.00%	81.40%	63.40%	0.70%	100.00%	0.0134
0.0025	1499477	280775	590	2151	84.20%	78.50%	84.20%	62.70%	0.80%	100.00%	0.0151
0.0030	1539300	240952	692	2049	86.40%	74.80%	86.50%	61.20%	0.80%	100.00%	0.0167
0.0035	1571817	208435	792	1949	88.30%	71.10%	88.30%	59.40%	0.90%	99.90%	0.0183
0.0040	1598330	181922	874	1867	89.70%	68.10%	89.80%	57.90%	1.00%	99.90%	0.0200
0.0045	1620295	159957	966	1775	91.00%	64.80%	91.00%	55.80%	1.10%	99.90%	0.0216
0.0050	1638739	141513	1051	1690	92.00%	61.70%	92.10%	53.70%	1.20%	99.90%	0.0232
0.0055	1654567	125685	1155	1586	92.90%	57.90%	92.90%	50.80%	1.20%	99.90%	0.0244
0.0060	1668438	111814	1229	1512	93.70%	55.20%	93.70%	48.90%	1.30%	99.90%	0.0261
0.0065	1680337	99915	1296	1445	94.30%	52.70%	94.40%	47.10%	1.40%	99.90%	0.0278
0.0070	1690808	89444	1376	1365	94.90%	49.80%	95.00%	44.80%	1.50%	99.90%	0.0292
0.0075	1699886	80366	1444	1297	95.40%	47.30%	95.50%	42.80%	1.60%	99.90%	0.0307
0.0080	1707777	72475	1505	1236	95.90%	45.10%	95.90%	41.00%	1.70%	99.90%	0.0323
0.0085	1714729	65523	1563	1178	96.20%	43.00%	96.30%	39.30%	1.80%	99.90%	0.0339
0.0090	1720783	59469	1640	1101	96.60%	40.20%	96.70%	36.80%	1.80%	99.90%	0.0348
0.0095	1726157	54095	1712	1029	96.90%	37.50%	97.00%	34.50%	1.90%	99.90%	0.0356
0.0100	1730957	49295	1762	979	97.10%	35.70%	97.20%	32.90%	1.90%	99.90%	0.0369

NC: Not calculated due to zero for either true positive or true negative.

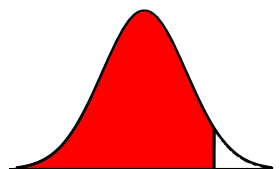
Bold indicates maximum values for each measure.



Model Assessment– Interpretation & Performance

$$\begin{aligned}\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = & -4.6829 + 0.3586X_5 - 2.1209X_{18} - 7.17 \times 10^{-8}X_1 \\ & - 6.43 \times 10^{-5}X_{10} - 2.6983 \times 10^{-2}X_{27} - 4.2024 \times 10^{-3}X_{40} \\ & + 9.74 \times 10^{-6}X_7 + 0.1061X_{32} - 0.7411X_{15} \\ & - 0.0297X_{29} - 1.50 \times 10^{-4}X_{11} - 0.3366X_{38} \\ & - 0.0167X_{26} - 1.16 \times 10^{-4}X_3 + 7.44 \times 10^{-4}X_{31} \\ & + 0.0108X_{21} + 0.0514X_{49}\end{aligned}$$

- **Accuracy** (at cutoff point 0.0020): **0.8140**
- **F1 Score** (at cutoff point 0.0020): **0.0134**





Towards Trustworthy Fraud Detection: Combining Class Imbalance Mitigation and Model Interpretability

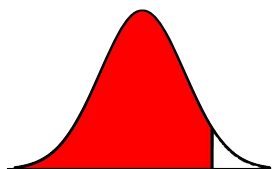
Student Name: Andy Chen, Alvin Lin, Chrissy Wang

Advisor Name: Hui Wen Teng

Appendix– candidates variables for Logistic Regression

Summary of variables

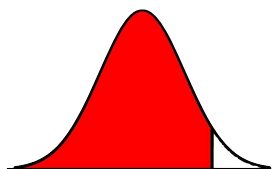
Index	Variables	Type	Index	Variables	Type
Y	Fraud_label	numeric	X ₁₄	zip_missing_flag	category
X ₁	transaction_id	numeric	X ₁₅	errors_missing_flag	category
X ₃	client_id	numeric	X ₁₆	use_chip_Chip Transaction	category
X ₄	card_id	numeric	X ₁₇	use_chip_Online Transaction	category
X ₅	amount	numeric	X ₁₈	use_chip_Swipe Transaction	category
X ₇	merchant_id	numeric	X ₁₉	fraud_label	category
X ₁₀	zip	numeric	X ₂₀	current_age	numeric
X ₁₁	mcc_code	numeric	X ₂₁	retirement_age	numeric
X ₁₃	merchant_state_missing_flag	category	X ₂₄	gender	category



Appendix– candidates variables for Logistic Regression

Summary of variables

Index	Variables	Type	Index	Variables	Type
X ₂₆	latitude	numeric	X ₃₉	num_cards_issued	numeric
X ₂₇	longitude	numeric	X ₄₀	credit_limit	numeric
X ₂₈	per_capita_income	numeric	X ₄₄	card_type_Credit	category
X ₂₉	yearly_income	numeric	X ₄₅	card_type_Debit	category
X ₃₀	total_debt	numeric	X ₄₆	card_type_Debit (Prepaid)	category
X ₃₁	credit_score	numeric	X ₄₇	card_brand_Amex	category
X ₃₂	num_credit_cards	numeric	X ₄₈	card_brand_Discover	category
X ₃₅	card_number	numeric	X ₄₉	card_brand_Mastercard	category
X ₃₇	cvv	numeric	X ₅₀	card_brand_Visa	category
X ₃₈	has_chip	category	X ₅₁	fraud_label_missing_flag	category



Andy 大綱草稿頁

1. **Why choose Logistics Regression as our Baseline model?**
(Property and assumptions– linearity of logit, independence of observations)
2. **Introduce independents variables and the dependent variables**
3. **To meet the multicollinearity assumption**
(a) – drop one hot encoding variables
(b) VIF index to check multicollinearity among variables
4. **Overview the variables we might choose (univariate tests of each independent variable)**
5. **Stepwise methods to choose variables**
6. **Diagnostics (ROC curve)**
7. **The final base model**

***Check linearity of logit (V)**

